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SUBBAND CONFIGURATION TECHNIQUES FOR SUBBAND FULL DUPLEX

Abstract

Methods, systems, and devices for wireless communications are described. A wireless device such as a user equipment (UE) may receive first information indicative of one or more bandwidth part (BWP) pairs configured for the UE, where each BWP pair includes a respective uplink BWP and a respective downlink BWP. The one or more BWP pairs may be active uplink and downlink BWPs that at least partially occupy a channel bandwidth of the UE. The UE may then receive second information indicative of a subband full duplex (SBFD) configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and are dedicated for the SBFD communications. The UE may then communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

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Background/Summary

CROSS REFERENCE [0001] The present application for patent claims benefit of U.S. Provisional Patent Application No. 63/557,384 by Abdelghaffar et al., entitled “Subband Configuration Techniques for Subband Full Duplex,” filed Feb. 23, 2024, assigned to the assignee hereof, and expressly incorporated herein.

FIELD OF TECHNOLOGY

[0002] The following relates to wireless communications, including subband configuration techniques for subband full duplex (SBFD).

BACKGROUND

[0003] Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations, each supporting wireless communication for communication devices, which may be known as user equipment (UE).

SUMMARY

[0004] The described techniques relate to improved methods, systems, devices, and apparatuses that support subband configuration techniques for subband full duplex (SBFD). For example, the described techniques provide for per-bandwidth part (BWP) or cell-specific SBFD configurations. A wireless device such as a user equipment (UE) may receive first information that indicates one or more BWP pairs configured for the UE, where each BWP pair includes a respective uplink BWP and a respective downlink BWP. The one or more BWP pairs may be active or initial uplink and downlink BWPs that at least partially occupy a channel bandwidth of the UE. The UE may then receive second information that indicates a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that correspond to each of the one or more BWP pairs, and are dedicated for the SBFD communications. The UE may then communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration. In some other implementations, the UE may receive a cell-specific SBFD configuration, based on a cell-specific channel bandwidth or a UE-specific channel bandwidth.

[0005] A method for wireless communications by a UE is described. The method may include receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and communicating one or more messages via the one or more BWP

pairs in accordance with the SBFD configuration.

[0006] A UE for wireless communications is described. The UE may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the UE to receive first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, receive second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0007] Another UE for wireless communications is described. The UE may include means for receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, means for receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0008] A non-transitory computer-readable medium storing code for wireless communications is described. The code may include instructions executable by one or more processors to receive first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, receive second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0009] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information may include operations, features, means, or instructions for transmitting an indication of one or more capabilities of the UE and receiving, responsive to the indication, the second information indicative of the SBFD configuration, where the SBFD configuration may be based on the one or more capabilities of the UE.

[0010] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the SBFD configuration includes a resource indication value (RIV) that indicates a resource allocation associated with the uplink subband, the downlink subband or both and the RIV includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation.

[0011] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the starting resource block may be indicated relative to a first physical resource block (PRB) of a carrier bandwidth.

[0012] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the starting resource block may be indicated relative to a first PRB of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE.

[0013] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the uplink subband, the downlink subband, or both, may be configured in an active BWP pair of the one or more BWP pairs.

[0014] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the SBFD configuration may be based on one or more capabilities of the UE, the one or more capabilities including a supported guard band size between the uplink subband and the

downlink subband, a threshold size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.

[0015] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, where a resource block group size of the resource allocation may be based on a size of the uplink subband, a size of the downlink subband, or both.

[0016] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, where the resource allocation of the scheduled physical channel includes a RIV that includes an indication of a starting resource block of the resource allocation that may be with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.

[0017] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, where the physical channel resource allocation indicates a starting resource block of each hop of the frequency hopping relative to a first resource block of the uplink subband, and where a resource block offset may be based on a size of the uplink subband, the downlink subband, or both.

[0018] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving the second information indicative of the SBFD configuration in respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, where the SBFD configuration may be indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration.

[0019] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more RIVs, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

[0020] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the SBFD configuration including the uplink subband, the downlink subband or both, may be configured in accordance with a common resource block grid of a cell configured for SBFD communications.

[0021] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the cell includes a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

[0022] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, communicating the one or more messages may include operations, features, means, or instructions for communicating the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based on one or more PRBs that at least partially overlap with the respective uplink BWP, the respective downlink BWP, or both, of the SBFD configuration.

[0023] Some examples of the method, UEs, and non-transitory computer-readable medium

described herein may further include operations, features, means, or instructions for dropping one or more PRBs of the respective uplink BWP, the respective downlink BWP, or both, based on a subcarrier spacing of the respective uplink BWP, the respective downlink BWP, or both, being greater than a corresponding subcarrier spacing of the uplink subband, the downlink subband, or both.

[0024] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving, while the UE may be in a connected state, the second information indicative of the SBFD configuration via a serving cell configuration message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0025] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving, while the UE may be in a connected state, the second information indicative of the SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0026] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving, while the UE may be in a connected state, an idle state, or an inactive state, the second information indicative of the SBFD configuration via a cell common configuration of a broadcast message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0027] Some examples of the method, UEs, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving the SBFD configuration includes a SBFD pattern periodicity that may be based on a time division duplexing (TDD) downlink and uplink periodicity.

[0028] In some examples of the method, UEs, and non-transitory computer-readable medium described herein, the UE may be operating in an inactive state or an idle state, and receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving the second information indicative of the SBFD configuration via one or more broadcast system information messages, where the uplink subband, the downlink subband, or both, may be configured relative to an initial BWP, may be configured based on a serving cell bandwidth, or both.

[0029] A method for wireless communications by a network entity is described. The method may include outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0030] A network entity for wireless communications is described. The network entity may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the network entity to output first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, output second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a

downlink subband or both, that corresponds to each of the one or more BWP pairs, and communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0031] Another network entity for wireless communications is described. The network entity may include means for outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, means for outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0032] A non-transitory computer-readable medium storing code for wireless communications is described. The code may include instructions executable by one or more processors to output first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE, output second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, and communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0033] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information may include operations, features, means, or instructions for obtaining an indication of one or more capabilities of the UE and outputting, responsive to the indication, the second information indicative of the SBFD configuration, where the SBFD configuration may be based on the one or more capabilities of the UE.

[0034] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the SBFD configuration includes a RIV that indicates a resource allocation associated with the uplink subband, the downlink subband or both and the RIV includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation.

[0035] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the starting resource block may be indicated relative to a first PRB of a carrier bandwidth.

[0036] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the starting resource block may be indicated relative to a first PRB of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE.

[0037] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the uplink subband, the downlink subband, or both, may be configured in an active BWP pair of the one or more BWP pairs.

[0038] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the SBFD configuration may be based on one or more capabilities of the UE, the one or more capabilities including a supported guard band size between the uplink subband and the downlink subband, a threshold size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.

[0039] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration may include operations, features, means, or instructions for outputting an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, where a resource block group size of the resource allocation may be based on a size of the uplink subband,

a size of the downlink subband, or both.

[0040] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration may include operations, features, means, or instructions for receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, where the resource allocation of the scheduled physical channel includes a RIV that includes an indication of a starting resource block of the resource allocation that may be with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.

[0041] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration may include operations, features, means, or instructions for outputting an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, where the physical channel resource allocation indicates a starting resource block of each hop of the frequency hopping relative to a first resource block of the uplink subband, and where a resource block offset may be based on a size of the uplink subband, the downlink subband, or both.

[0042] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration may include operations, features, means, or instructions for outputting the second information indicative of the SBFD configuration in respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, where the SBFD configuration may be indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration.

[0043] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more RIVs, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

[0044] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the SBFD configuration including the uplink subband, the downlink subband or both, may be configured in accordance with a common resource block grid of a cell configured for SBFD communications.

[0045] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, the cell includes a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

[0046] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, communicating the one or more messages may include operations, features, means, or instructions for communicating the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based on one or more PRBs that at least partially overlap with the respective uplink BWP, the respective downlink BWP, or both, of the SBFD configuration.

[0047] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration may include operations, features, means, or instructions for outputting, while the UE may be in a connected state, the second information indicative of the SBFD configuration via a serving cell configuration message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0048] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration

may include operations, features, means, or instructions for outputting, while the UE may be in a connected state, the second information indicative of the SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0049] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, receiving the second information indicative of the SBFD configuration may include operations, features, means, or instructions for outputting, while the UE may be in a connected state, an idle state, or an inactive state, the second information indicative of the SBFD configuration via a cell common configuration of a broadcast message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0050] Some examples of the method, network entities, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for outputting the SBFD configuration includes a SBFD pattern periodicity that may be based on a TDD downlink and uplink periodicity.

[0051] In some examples of the method, network entities, and non-transitory computer-readable medium described herein, outputting the second information indicative of the SBFD configuration may include operations, features, means, or instructions for outputting the second information indicative of the SBFD configuration via one or more broadcast system information messages, where the uplink subband, the downlink subband, or both, may be configured relative to an initial BWP, may be configured based on a serving cell bandwidth, or both.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0052] FIGS. **1** and **2** show examples of wireless communications systems that support subband configuration techniques for subband full duplex (SBFD) in accordance with one or more aspects of the present disclosure.

[0053] FIGS. **3**, **4**, **5**, and **6** show examples of SBFD configurations that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0054] FIG. **7** shows an example of a wireless communications system that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0055] FIGS. **8**, **9**, and **10** show examples of SBFD configurations that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0056] FIG. **11** shows an example of a process flow that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0057] FIGS. **12** and **13** show block diagrams of devices that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0058] FIG. **14** shows a block diagram of a communications manager that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0059] FIG. **15** shows a diagram of a system including a device that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0060] FIGS. **16** and **17** show block diagrams of devices that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0061] FIG. **18** shows a block diagram of a communications manager that supports subband

configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0062] FIG. 19 shows a diagram of a system including a device that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

[0063] FIGS. 20 through 22 show flowcharts illustrating methods that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0064] Some wireless communications systems may support subband full duplex (SBFD) communications, where a carrier (such as a cell-specific carrier or a user equipment (UE)-specific carrier) may be partitioned into one or more subbands for downlink communications, uplink communications, or both. In some cases, SBFD techniques may allow downlink and uplink communications on corresponding downlink and uplink subbands to occur simultaneously (e.g., at the same time) in a same slot. For example, a network entity may allocate a set of frequency resources (such as a set of dedicated uplink and downlink subbands) within a channel, which may be used by devices to perform SBFD uplink and downlink communications. In some implementations, the SBFD communications may occur within one or more active bandwidth parts (BWPs) configured for a UE, such that the UE may receive downlink communications using downlink subband frequency resources within an active downlink BWP, and may transmit uplink communications using corresponding uplink subband frequency resources in an active uplink BWP. In some cases, however, the UE may be unaware of which uplink and downlink resources to use for SBFD communications, including uplink and downlink subbands within one or more active BWPs, or which resources out of a set of indicated resources are available to use for SBFD.

[0065] To support efficient SBFD resource allocation and communication within a wireless communications system, the network entity may configure an uplink and downlink SBFD configuration for each uplink and downlink BWP pair (including at least one uplink BWP and one downlink BWP) configured for the UE, or a cell-specific uplink and downlink SBFD configuration. For example, the uplink and downlink SBFD configuration may be indicative of a resource allocation associated with SBFD communications for the UE per uplink and downlink BWP pair, where the resource allocation indicates different dedicated SBFD uplink and downlink subbands that correspond to each uplink and downlink BWP within the channel bandwidth of the UE.

[0066] In some cases, the SBFD configuration may be based on (or configured to support) one or more capabilities of the UE, may be based on an operating mode of the UE (including radio resource control (RRC) connected modes and idle or inactive modes of the UE), may be based on UE guard band size or configured subband size, among other supported UE capabilities. In some aspects, the uplink and downlink SBFD configuration may allow the UE to efficiently determine a frequency location and associated configuration for the one or more uplink and downlink subbands in each active uplink or downlink BWP.

[0067] The techniques described herein may be implemented to realize one or more potential advantages. For example, SBFD techniques, when implemented by the network and the UE, may increase system throughput based on increased opportunities to transmit and receive uplink and downlink communications (e.g., simultaneous uplink and downlink communications). Additionally, or alternatively, the techniques described herein may increase uplink and downlink coverage and signaling reliability for the UE, and may reduce latency of both uplink and downlink communications based on the SBFD configuration, where the UE may be aware of dedicated uplink and downlink subbands for SBFD communications within each active BWP allocated to the UE.

[0068] Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects of the disclosure are further illustrated by and described with reference to SBFD

configurations, a process flow, apparatus diagrams, system diagrams, and flowcharts that relate to subband configuration techniques for SBFD.

[0069] FIG. 1 shows an example of a wireless communications system **100** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The wireless communications system **100** may include one or more devices, such as one or more network devices (e.g., network entities **105**), one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0070] The network entities **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may include devices in different forms or having different capabilities. In various examples, a network entity **105** may be referred to as a network element, a mobility element, a radio access network (RAN) node, or network equipment, among other nomenclature. In some examples, network entities **105** and UEs **115** may wirelessly communicate via communication link(s) **125** (e.g., a radio frequency (RF) access link). For example, a network entity **105** may support a coverage area **110** (e.g., a geographic coverage area) over which the UEs **115** and the network entity **105** may establish the communication link(s) **125**. The coverage area **110** may be an example of a geographic area over which a network entity **105** and a UE **115** may support the communication of signals according to one or more radio access technologies (RATs).

[0071] The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. 1. The UEs **115** described herein may be capable of supporting communications with various types of devices in the wireless communications system **100** (e.g., other wireless communication devices, including UEs **115** or network entities **105**), as shown in FIG. 1.

[0072] As described herein, a node of the wireless communications system **100**, which may be referred to as a network node, or a wireless node, may be a network entity **105** (e.g., any network entity described herein), a UE **115** (e.g., any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE **115**. As another example, a node may be a network entity **105**. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a UE **115**. In another aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a network entity **105**. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE **115**, network entity **105**, apparatus, device, computing system, or the like may include disclosure of the UE **115**, network entity **105**, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE **115** is configured to receive information from a network entity **105** also discloses that a first node is configured to receive information from a second node.

[0073] In some examples, network entities **105** may communicate with a core network **130**, or with one another, or both. For example, network entities **105** may communicate with the core network **130** via backhaul communication link(s) **120** (e.g., in accordance with an S1, N2, N3, or other interface protocol). In some examples, network entities **105** may communicate with one another via backhaul communication link(s) **120** (e.g., in accordance with an X2, Xn, or other interface protocol) either directly (e.g., directly between network entities **105**) or indirectly (e.g., via the core

network **130**). In some examples, network entities **105** may communicate with one another via a midhaul communication link **162** (e.g., in accordance with a midhaul interface protocol) or a fronthaul communication link **168** (e.g., in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication link(s) **120**, midhaul communication links **162**, or fronthaul communication links **168** may be or include one or more wired links (e.g., an electrical link, an optical fiber link) or one or more wireless links (e.g., a radio link, a wireless optical link), among other examples or various combinations thereof. A UE **115** may communicate with the core network **130** via a communication link **155**.

[0074] One or more of the network entities **105** or network equipment described herein may include or may be referred to as a base station **140** (e.g., a base transceiver station, a radio base station, an NR base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some examples, a network entity **105** (e.g., a base station **140**) may be implemented in an aggregated (e.g., monolithic, standalone) base station architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within one network entity (e.g., a network entity **105** or a single RAN node, such as a base station **140**).

[0075] In some examples, a network entity **105** may be implemented in a disaggregated architecture (e.g., a disaggregated base station architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among multiple network entities (e.g., network entities **105**), such as an integrated access and backhaul (IAB) network, an open RAN (O-RAN) (e.g., a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (e.g., a cloud RAN (C-RAN)). For example, a network entity **105** may include one or more of a central unit (CU), such as a CU **160**, a distributed unit (DU), such as a DU **165**, a radio unit (RU), such as an RU **170**, a RAN Intelligent Controller (RIC), such as an RIC **175** (e.g., a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) system, such as an SMO system **180**, or any combination thereof. An RU **170** may also be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (e.g., separate physical locations). In some examples, one or more of the network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (e.g., a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0076] The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending on which functions (e.g., network layer functions, protocol layer functions, baseband functions, RF functions, or any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some examples, the CU **160** may host upper protocol layer (e.g., layer 3 (L3), layer 2 (L2)) functionality and signaling (e.g., Radio Resource Control (RRC), service data adaptation protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** (e.g., one or more CUs) may be connected to a DU **165** (e.g., one or more DUs) or an RU **170** (e.g., one or more RUs), or some combination thereof, and the DUs **165**, RUs **170**, or both may host lower protocol layers, such as layer 1 (L1) (e.g., physical (PHY) layer) or L2 (e.g., radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the

protocol stack. The DU **165** may support one or multiple different cells (e.g., via one or multiple different RUs, such as an RU **170**). In some cases, a functional split between a CU **160** and a DU **165** or between a DU **165** and an RU **170** may be within a protocol layer (e.g., some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to a DU **165** via a midhaul communication link **162** (e.g., F1, F1-c, F1-u), and a DU **165** may be connected to an RU **170** via a fronthaul communication link **168** (e.g., open fronthaul (FH) interface). In some examples, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (e.g., a channel) between layers of a protocol stack supported by respective network entities (e.g., one or more of the network entities **105**) that are in communication via such communication links.

[0077] In some wireless communications systems (e.g., the wireless communications system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (e.g., to a core network **130**). In some cases, in an IAB network, one or more of the network entities **105** (e.g., network entities **105** or IAB node(s) **104**) may be partially controlled by each other. The IAB node(s) **104** may be referred to as a donor entity or an IAB donor. A DU **165** or an RU **170** may be partially controlled by a CU **160** associated with a network entity **105** or base station **140** (such as a donor network entity or a donor base station). The one or more donor entities (e.g., IAB donors) may be in communication with one or more additional devices (e.g., IAB node(s) **104**) via supported access and backhaul links (e.g., backhaul communication link(s) **120**). IAB node(s) **104** may include an IAB mobile termination (IAB-MT) controlled (e.g., scheduled) by one or more DUs (e.g., DUs **165**) of a coupled IAB donor. An IAB-MT may be equipped with an independent set of antennas for relay of communications with UEs **115** or may share the same antennas (e.g., of an RU **170**) of IAB node(s) **104** used for access via the DU **165** of the IAB node(s) **104** (e.g., referred to as virtual IAB-MT (vIAB-MT)). In some examples, the IAB node(s) **104** may include one or more DUs (e.g., DUs **165**) that support communication links with additional entities (e.g., IAB node(s) **104**, UEs **115**) within the relay chain or configuration of the access network (e.g., downstream). In such cases, one or more components of the disaggregated RAN architecture (e.g., the IAB node(s) **104** or components of the IAB node(s) **104**) may be configured to operate according to the techniques described herein.

[0078] In the case of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support test as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (e.g., a base station **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (e.g., components such as an IAB node, a DU **165**, a CU **160**, an RU **170**, an RIC **175**, an SMO system **180**).

[0079] A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, vehicles, or meters, among other examples.

[0080] The UEs **115** described herein may be able to communicate with various types of devices, such as UEs **115** that may sometimes operate as relays, as well as the network entities **105** and the

network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. 1.

[0081] The UEs **115** and the network entities **105** may wirelessly communicate with one another via the communication link(s) **125** (e.g., one or more access links) using resources associated with one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined PHY layer structure for supporting the communication link(s) **125**. For example, a carrier used for the communication link(s) **125** may include a portion of an RF spectrum band (e.g., a BWP (BWP)) that is operated according to one or more PHY layer channels for a given RAT (e.g., LTE, LTE-A, LTE-A Pro, NR). Each PHY layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (e.g., entity, sub-entity) of a network entity **105**. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity **105**, may refer to any portion of a network entity **105** (e.g., a base station **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (e.g., directly or via one or more other network entities, such as one or more of the network entities **105**).

[0082] In some examples, such as in a carrier aggregation configuration, a carrier may have acquisition signaling or control signaling that coordinates operations for other carriers. A carrier may be associated with a frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute RF channel number (EARFCN)) and may be identified according to a channel raster for discovery by the UEs **115**. A carrier may be operated in a standalone mode, in which case initial acquisition and connection may be conducted by the UEs **115** via the carrier, or the carrier may be operated in a non-standalone mode, in which case a connection is anchored using a different carrier (e.g., of the same or a different RAT).

[0083] The communication link(s) **125** of the wireless communications system **100** may include downlink transmissions (e.g., forward link transmissions) from a network entity **105** to a UE **115**, uplink transmissions (e.g., return link transmissions) from a UE **115** to a network entity **105**, or both, among other configurations of transmissions. Carriers may carry downlink or uplink communications (e.g., in an FDD mode) or may be configured to carry downlink and uplink communications (e.g., in a TDD mode).

[0084] A carrier may be associated with a particular bandwidth of the RF spectrum, and, in some examples, the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a set of bandwidths for carriers of a particular RAT (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communications system **100** (e.g., the network entities **105**, the UEs **115**, or both) may have hardware configurations that support communications using a particular carrier bandwidth or may be configurable to support communications using one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include network entities **105** or UEs **115** that support concurrent communications using carriers associated with multiple carrier bandwidths. In some examples, each served UE **115** may be configured for operating using portions (e.g., a sub-band, a BWP) or all of a carrier bandwidth.

[0085] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (e.g.,

a duration of one modulation symbol) and one subcarrier, in which case the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (e.g., in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (e.g., a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.

[0086] One or more numerologies for a carrier may be supported, and a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE **115** may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and communications for the UE **115** may be restricted to one or more active BWPs.

[0087] The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of $T_{\text{sub.s}} = 1/(\Delta f_{\text{sub.max}} \cdot N_{\text{sub.f}})$ seconds, for which $\Delta f_{\text{sub.max}}$ may represent a supported subcarrier spacing, and $N_{\text{sub.f}}$ may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).

[0088] Each frame may include multiple consecutively-numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems, such as the wireless communications system **100**, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (e.g., $N_{\text{sub.f}}$) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation. [0089] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs)).

[0090] Physical channels may be multiplexed for communication using a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given

payload size. Search space sets may include common search space sets configured for sending control information to UEs **115** (e.g., one or more UEs) or may include UE-specific search space sets for sending control information to a UE **115** (e.g., a specific UE).

[0091] A network entity **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a network entity **105** (e.g., using a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID)). In some examples, a cell also may refer to a coverage area **110** or a portion of a coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the network entity **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with coverage areas **110**, among other examples.

[0092] A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a network entity **105** operating with lower power (e.g., a base station **140** operating with lower power) relative to a macro cell, and a small cell may operate using the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A network entity **105** may support one or more cells and may also support communications via the one or more cells using one or multiple component carriers.

[0093] In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.

[0094] In some examples, a network entity **105** (e.g., a base station **140**, an RU **170**) may be movable and therefore provide communication coverage for a moving coverage area, such as the coverage area **110**. In some examples, coverage areas **110** (e.g., different coverage areas) associated with different technologies may overlap, but the coverage areas **110** (e.g., different coverage areas) may be supported by the same network entity (e.g., a network entity **105**). In some other examples, overlapping coverage areas, such as a coverage area **110**, associated with different technologies may be supported by different network entities (e.g., the network entities **105**). The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the network entities **105** support communications for coverage areas **110** (e.g., different coverage areas) using the same or different RATs.

[0095] The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

[0096] In some examples, a UE **115** may be configured to support communicating directly with other UEs (e.g., one or more of the UEs **115**) via a device-to-device (D2D) communication link, such as a D2D communication link **135** (e.g., in accordance with a peer-to-peer (P2P), D2D, or

sidelink protocol). In some examples, one or more UEs **115** of a group that are performing D2D communications may be within the coverage area **110** of a network entity **105** (e.g., a base station **140**, an RU **170**), which may support aspects of such D2D communications being configured by (e.g., scheduled by) the network entity **105**. In some examples, one or more UEs **115** of such a group may be outside the coverage area **110** of a network entity **105** or may be otherwise unable to or not configured to receive transmissions from a network entity **105**. In some examples, groups of the UEs **115** communicating via D2D communications may support a one-to-many (1:M) system in which each UE **115** transmits to one or more of the UEs **115** in the group. In some examples, a network entity **105** may facilitate the scheduling of resources for D2D communications. In some other examples, D2D communications may be carried out between the UEs **115** without an involvement of a network entity **105**.

[0097] The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the network entities **105** (e.g., base stations **140**) associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0098] The wireless communications system **100** may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. Communications using UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than one hundred kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0099] The wireless communications system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) RAT, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations using unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (e.g., LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0100] A network entity **105** (e.g., a base station **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example,

one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a network entity **105** may be located at diverse geographic locations. A network entity **105** may include an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0101] Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a network entity **105**, a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0102] In some examples, a UE **115** and a network entity **105** may operate using a set of frequency resources of a frequency band (e.g., FR1). For example, a carrier or operator may be allocated a subset of a total operating band, and the channel bandwidth of the carrier may be allocated as a cell-specific bandwidth of a cell associated with the network entity **105**. In some examples, the UE **115** may be allocated a set of resources of the cell-specific bandwidth, where the UE-specific channel bandwidth may be different from or the same as the cell-specific bandwidth. For example, the UE **115** may support a maximum bandwidth of 60 MHz (or another maximum bandwidth) while the network entity **105** may support a channel bandwidth that is 100 MHz (or another maximum bandwidth that is greater than or equal to the UE bandwidth). In such examples, the UE **115** may be assigned or configured to operate in an operating bandwidth of up to 60 MHz that is a portion or subset of the channel bandwidth of the network entity **105**.

[0103] In some implementations, the UE-specific channel bandwidth (and channel allocation) may be configured in a serving cell configuration (e.g., as ‘downlinkChannelBW-PerSCS-List’ and ‘uplinkChannelBW-PerSCS-List’ of the ‘servingCellConfig’) after the UE **115** performs RRC connection with the cell. The UE **115** may set the channel bandwidth to operate in compliance with various communications regulations. Additionally, or alternatively, the cell-specific carrier bandwidth and carrier location may be indicated via one or more parameters (e.g., SCS-SpecificCarrier) that may be included in one or more broadcast messages (e.g., SIB1). The UE **115** may use the information in the one or more broadcast messages to determine where the PRBs are located in relation to a reference point, and whether the BWP configuration is intended for the UE **115** (or for one or more other UEs in the network).

[0104] Wireless communications system **100** may support SBFD communications, where an allocated frequency band may be partitioned into one or more subbands for downlink communications, uplink communications, or both. In some cases, SBFD techniques may allow downlink and uplink communications on corresponding downlink and uplink subbands to occur simultaneously. In some implementations, the SBFD communications may occur within one or more active BWPs configured for a UE **115**, where the UE **115** (which may have SBFD capabilities such as half duplex capabilities, full duplex capabilities, or both) may receive downlink communications using downlink subband frequency resources within an active downlink BWP, and

may transmit uplink communications using corresponding uplink subband frequency resources in an active uplink BWP. In some cases, however, the UE **115** may be unaware of which uplink and downlink resources to use for SBFD communications, including uplink and downlink subbands within one or more active or initial BWPs, or which resources out of a set of indicated resources are available to use for SBFD.

[0105] To support efficient SBFD resource allocation and communication within a wireless communications system, the network entity may configure an uplink and downlink SBFD configuration for each uplink and downlink BWP pair (including at least one uplink BWP and one downlink BWP) configured for the UE **115**, or a cell-specific uplink and downlink SBFD configuration. For example, the uplink and downlink SBFD configuration may indicate different dedicated SBFD uplink and downlink subbands that correspond to each uplink and downlink BWP within the channel bandwidth of the UE **115**.

[0106] FIG. **2** shows an example of a wireless communications system **200** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the wireless communications system **200** may illustrate communications between a network entity **105** and a UE **115**, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIG. **1**. In some examples, the UE **115** and network entity **105** may support SBFD communications including uplink and downlink signaling in an allocated frequency band.

[0107] For example, the wireless communications system **200** may support SBFD communications in which a single frequency band may be partitioned into one or more sub-bands for downlink communications, uplink communications, or both, which may occur simultaneously (e.g., at the same time, using one or more same time resources) in a same slot. Some SBFD techniques, when implemented by the network entity **105** and the UE **115**, may increase system throughput, increase uplink and downlink coverage for the UE **115**, and may reduce latency of both uplink and downlink communications.

[0108] The network entity **105** may allocate a set of frequencies within a channel bandwidth (or a component carrier bandwidth) into one or more downlink subbands, one or more uplink subbands, and one or more corresponding guard bands in between the one or more downlink subbands and the one or more uplink subbands. For example, for an SBFD configuration including the one or more downlink subbands, the one or more uplink subbands and the one or more guard bands may indicate a non-overlapping SBFD configuration. In some aspects, SBFD communications may occur within one or more active BWPs, where the UE **115** may receive downlink communications using downlink subband frequency resources within an active downlink BWP, and may transmit uplink communications using corresponding uplink subband frequency resources in an active uplink BWP. In some implementations, however, the UE **115** may be unaware of which uplink and downlink resources to use for SBFD communications, including uplink and downlink subbands within one or more active BWPs, or which resources out of a set of indicated resources are available to use for SBFD.

[0109] In some examples, the network entity **105** may signal or otherwise indicate an uplink and downlink SBFD configuration as a single cell-common configuration, or using a per-BWP configuration. For the single cell-common configuration, the downlink and uplink subbands may be defined within a channel bandwidth of the carrier (e.g., either a cell-specific network carrier bandwidth, or a UE-specific carrier bandwidth, where the network carrier bandwidth and the UE-specific carrier bandwidth may be different). For an SBFD configuration per-BWP, the network entity **105** may convey multiple SBFD configurations for each BWP allocated for the UE **115**, for example, the UE **115** may have one or more dedicated downlink and uplink subbands for each corresponding downlink and uplink BWP configured for the UE **115**. In such examples, the downlink and uplink subbands may be located within the active uplink and downlink BWPs of the UE **115**, such that the UE **115** may utilize the uplink and downlink resources within the active

uplink and downlink BWPs.

[0110] To support efficient SBFD resource allocation and communication within a wireless communications system, the network entity **105** may configure the uplink and downlink SBFD configuration **205** for each uplink and downlink BWP pair (including uplink BWP **225** and downlink BWP **230**) configured for the UE **115**. In some implementations, the uplink and downlink SBFD configuration **205** may be indicative of a resource allocation **210** associated with SBFD communications for the UE **115**. The network entity **105** may support a cell bandwidth that includes a cell downlink subband **215-a**, a cell uplink subband **220**, and a cell downlink subband **215-b**. The UE **115** may be configured with one or more uplink and downlink BWP pairs (one of which may include uplink BWP **225** and downlink BWP **230**) within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands (e.g., uplink subband **235** and downlink subband **240**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**. For example, the UE **115** may have a dedicated set of subbands, including uplink subband **235** and downlink subband **240**, that may span the same frequency as the uplink and downlink BWP pair. In some cases, the dedicated set of subbands may correspond to the set of cell-specific subbands. For example, the uplink BWP **225** overlaps with the cell uplink subband **220**, and the corresponding SBFD uplink subband for the UE **115** includes the overlapped portion of the uplink BWP **225** and the cell uplink subband **220**. In addition, the downlink BWP **230** overlaps with the cell downlink subband **215-b**, so the downlink subband **240** (e.g., a corresponding SBFD downlink subband) includes the overlapped portion of the downlink BWP **230** and the cell downlink subband **215-b**.

[0111] In some cases, the SBFD configuration may be based on (or configured to support) one or more capabilities of the UE **115**, may be configured based on an operating mode of the UE **115** (including RRC connected modes and idle or inactive modes of the UE **115**), UE required guard band size, subband size requirements, among other supported UE capabilities. The uplink and downlink SBFD configuration **205** may allow the UE **115** to efficiently determine a frequency location and associated configuration for the one or more uplink and downlink subbands in each allocated uplink or downlink BWP.

[0112] FIG. **3** shows an example of SBFD configuration **301** and SBFD configuration **302** that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the SBFD configuration **301** and the SBFD configuration **302** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. **1** and **2**. For example, the SBFD configuration **301** and the SBFD configuration **302** may be configured by a network entity **105**, and may be implemented by a UE **115**, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. **1** and **2**. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0113] In some examples of the SBFD configuration **301** and the SBFD configuration **302**, the network entity **105** may configure both uplink and downlink SBFD communications for each BWP pair supported by the UE **115**. For example, the SBFD configuration **301** may illustrate an example BWP pair configuration including uplink BWP **315-a** and downlink BWP **320-a**, which may be configured for the UE **115** within (at least) a portion of the cell bandwidth supported by the network entity **105**, including a first cell downlink subband **305-a**, a cell uplink subband **310-a**, and a second cell downlink subband **305-b**, which may be separated from one another by a cell guard band.

[0114] In some examples, the UE **115** may operate within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands (e.g., dedicated

uplink subband **325-a** and dedicated downlink subband **330-a**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**. For example, the UE **115** may have a dedicated set of subbands, including dedicated uplink subband **325-a** and dedicated downlink subband **330-a** that may span the same frequency as the uplink and downlink BWP pair. In some cases, the set of dedicated subbands (e.g., dedicated uplink subband **325-a** and dedicated downlink subband **330-a**) may correspond to the set of cell-specific subbands. For example, the uplink BWP **315-a** (which may be an active uplink BWP for the UE **115**) may overlap with the cell uplink subband **310-a**, and the corresponding SBFD uplink subband for the UE **115** (e.g., dedicated uplink subband **325-a**) includes uplink resources located within the overlapped portion of the uplink BWP **315-a** and the cell uplink subband **310-a**. In addition, the downlink BWP **320-a** (which may be an active downlink BWP for the UE **115**) overlaps with the second cell downlink subband **305-b**, so the dedicated downlink subband **330-a** includes resources of the overlapped portion of the downlink BWP **320-a** and the second cell downlink subband **305-b**. In some examples, since the dedicated downlink subband **330-a** and the dedicated uplink subband **325-a** are located within the active uplink and downlink BWPs for the UE **115**, the UE **115** may identify resources of the dedicated subbands as usable resources (e.g., resources allocated for the UE **115** that are available or otherwise usable by the UE **115**). Additionally, or alternatively, the dedicated downlink subband **330-a** and the dedicated uplink subband **325-a** may be separated by a UE dedicated guard band, which may be configured based on various parameters, such as one or more UE capabilities.

[0115] The SBFD configuration **302** may illustrate another example BWP pair configuration for the UE **115** that includes uplink BWP **315-b** and downlink BWP **320-b**, which may be configured for the UE **115** within (at least) a portion of the cell bandwidth supported by the network entity **105**, including a first cell downlink subband **305-c**, a cell uplink subband **310-b**, and a second cell downlink subband **305-d**, which may be separated from one another by a cell guard band.

[0116] In some examples, the UE **115** may operate within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands per BWP (e.g., dedicated uplink subband **325-b**, a first dedicated downlink subband **330-b**, and a second dedicated downlink subband **330-c**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**. For example, the UE **115** may be configured a dedicated set of subbands, including dedicated uplink subband **325-b**, the first dedicated downlink subband **330-b**, and the second dedicated downlink subband **330-c**, that may span the same frequency as the uplink and downlink BWP pair. In some cases, the set of dedicated subbands (e.g., dedicated uplink subband **325-b**, the first dedicated downlink subband **330-b**, and the second dedicated downlink subband **330-c**) may correspond to the set of cell-specific subbands. For example, the uplink BWP **315-b** (which may be an active uplink BWP for the UE **115**) may fully overlap with the cell uplink subband **310-b**, and the corresponding SBFD uplink subband for the UE **115** (e.g., dedicated uplink subband **325-b**) includes uplink resources located within the overlapped portion of the uplink BWP **315-b** and the cell uplink subband **310-b**. In addition, the downlink BWP **320-b** (which may be an active downlink BWP for the UE **115**) fully overlaps with the first dedicated downlink subband **330-b** and the second dedicated downlink subband **330-c**, so the first dedicated downlink subband **330-b** and second dedicated downlink subband **330-c** includes resources of the overlapped portion of the downlink BWP **320-b** and both the first cell downlink subband **305-c** and the second cell downlink subband **305-d**.

[0117] In some implementations, the configuration of the dedicated uplink and downlink subbands for SBFD may be based on one or more capabilities of the UE **115**. For example, the UE **115** may support up to a threshold guard band size between the dedicated uplink subband **325-a** and the dedicated downlink subbands (e.g., the first dedicated downlink subband **330-b** and the second dedicated downlink subband **330-c**). Additionally, or alternatively, the SBFD configuration may be

based on one or more supported uplink and downlink subband sizes (e.g., the UE **115** may support one or more required sizes of the dedicated uplink subband **325-a**, the first dedicated downlink subband **330-b**, and the second dedicated downlink subband **330-c**). In some examples, the supported uplink and downlink subband sizes may be configured to match an existing channel bandwidth of the UE, or may be configured to satisfy one or more size requirements (e.g., a minimum size or maximum size). In some aspects, the UE **115** may support capability signaling in which the UE **115** may transmit one or more messages that indicate one or more capabilities of the UE **115**. For example, the UE **115** may signal a bitmap that is indicative of one or more sizes of uplink subbands, downlink subbands, or both, that the UE **115** supports.

[0118] FIG. **4** shows an example of SBFD configuration **401** and SBFD configuration **402** that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the SBFD configuration **401** and the SBFD configuration **402** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. **1** and **2**. For example, the SBFD configuration **401** and the SBFD configuration **402** may be configured by a network entity **105**, and may be implemented by a UE **115**, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. **1** and **2**. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0119] In some examples of the SBFD configuration **401** and the SBFD configuration **402**, the network entity **105** may configure both uplink and downlink SBFD communications for each BWP pair supported by the UE **115**. For example, the SBFD configuration **401** may illustrate an example BWP pair configuration including uplink BWP **415-a** and downlink BWP **420-a**, which may be configured for the UE **115** within (at least) a portion of the cell bandwidth supported by the network entity **105**, including a first cell downlink subband **405-a**, a cell uplink subband **410-a**, and a second cell downlink subband **405-b**, which may be separated from one another by a cell guard band.

[0120] In some examples, the UE **115** may operate within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands (e.g., dedicated uplink subband **425**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**. For example, the UE **115** may have a dedicated subband, including dedicated uplink subband **425** that may span the same frequency as the uplink and downlink BWP pair. In some cases, the dedicated subband (e.g., dedicated uplink subband **425**) may correspond to the set of cell-specific subbands. For example, the uplink BWP **415-a** (which may be an active uplink BWP for the UE **115**) may overlap with the cell uplink subband **410-a**, and the corresponding SBFD uplink subband for the UE **115** (e.g., dedicated uplink subband **425**) may include uplink resources located within the overlapped portion of the uplink BWP **415-a** and the cell uplink subband **410-a**. In addition, the downlink BWP **420-a** (which may be an active downlink BWP for the UE **115**) also may overlap with the cell uplink subband **410-a**. In such examples, since the downlink BWP **420-a** is non-overlapping with the first cell downlink subband **405-a** or the second cell downlink subband **405-b**, the UE **115** may lack a corresponding dedicated downlink subband for the uplink and downlink BWP pair. For example, the UE **115**, for SBFD-configured symbols, may transmit uplink signals and uplink channels within its active uplink BWP (e.g., uplink BWP **415-a**), and may not receive downlink signals or downlink channels from the network entity **105** within its active downlink BWP (e.g., downlink BWP **420-a**). In such examples, the downlink BWP **420-a** may be non-overlapping with the first cell downlink subband **405-a** or the second cell downlink subband **405-b**, or both.

[0121] The SBFD configuration **402** may illustrate an example BWP pair configuration including uplink BWP **415-b** and downlink BWP **420-b**, which may be configured for the UE **115** within (at

least) a portion of the cell bandwidth supported by the network entity **105**, including a first cell downlink subband **405-c**, a cell uplink subband **410-b**, and a second cell downlink subband **405-d**, which may be separated from one another by a cell guard band.

[0122] In some examples, the UE **115** may operate within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands (e.g., dedicated downlink subband **430**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**. For example, the UE **115** may have a dedicated subband, including dedicated downlink subband **430** that may span the same frequency as the uplink and downlink BWP pair. In some cases, the dedicated subband (e.g., dedicated downlink subband **430**) may correspond to the set of cell-specific subbands. For example, the uplink BWP **415-b** (which may be an active uplink BWP for the UE **115**) may overlap with the second cell downlink subband **405-d**, and may not overlap with the cell uplink subband **410-b**. In such examples, the UE **115** may not have a corresponding uplink SBFD subband since the active uplink BWP does not overlap with the cell uplink subband **410-b**. The downlink BWP **420-b** may overlap with the second cell downlink subband **405-d**, such that the corresponding SBFD downlink subband for the UE **115** (e.g., dedicated downlink subband **430**) may include downlink resources located within the overlapped portion of the downlink BWP **420-b** and the second cell downlink subband **405-d**. For example, the UE **115**, for SBFD-configured symbols, may receive downlink signals and downlink channels within its active downlink BWP (e.g., downlink BWP **420-b**) and may not transmit uplink signals or uplink channels to the network entity **105** within its active uplink BWP (e.g., uplink BWP **415.b**), due to an absence of a UE dedicated UL subband, due to the uplink BWP **415-b** being non-overlapping with the cell uplink subband **410-b**.

[0123] In some implementations, the configuration of the dedicated uplink and downlink subbands for SBFD may be based on one or more capabilities of the UE **115**. For example, the SBFD configuration may be based on one or more supported uplink and downlink subband sizes (e.g., the UE **115** may support one or more required sizes of the dedicated uplink subband **425** and the dedicated downlink subband **430**). In some examples, the supported uplink and downlink subband sizes may be configured to match an existing channel bandwidth of the UE **115**, or may be configured to satisfy one or more size requirements (e.g., minimum size or maximum size). In some aspects, the UE **115** may support capability signaling in which the UE **115** may transmit one or more messages that indicate one or more capabilities of the UE **115**. For example, the UE **115** may signal a bitmap that is indicative of one or more sizes of uplink subbands, downlink subbands, or both, that the UE **115** supports.

[0124] FIG. 5 shows an example of a SBFD configuration **501** and an SBFD configuration **502** that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the SBFD configuration **501** and the SBFD configuration **502** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. 1 and 2. For example, the SBFD configuration **501** and the SBFD configuration **502** may be configured by a network entity **105**, and may be implemented by a UE **115**, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. 1 and 2. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0125] In some examples of the SBFD configuration **501** and the SBFD configuration **502**, the network entity **105** may configure both uplink and downlink SBFD communications for each BWP pair supported by the UE **115**. For example, the SBFD configuration **501** may illustrate an example BWP pair configuration including uplink BWP **515-a** and downlink BWP **520-a**, which may be configured for the UE **115** within (at least) a portion of the cell bandwidth supported by the network entity **105**, including a first cell downlink subband **505-a**, a cell uplink subband **510-a**, and

a second cell downlink subband **505-b**, which may be separated from one another by a cell guard band. In some examples, the uplink BWP **515-a** and the downlink BWP **520-a** may span different bandwidths (e.g., the downlink BWP **520-a** may be relatively larger than the uplink BWP **515-a** to support increased downlink throughput, urgent traffic, or to support a relatively large quantity of downlink communications that exceeds a traffic threshold).

[0126] In some examples, the UE **115** may operate within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands (e.g., first dedicated downlink subband **525-a**, second dedicated downlink subband **525-b**, and dedicated uplink subband **530-a**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**, and which are separated by a UE dedicated guard band. For example, the UE **115** may have dedicated subbands for SBFD, including the first dedicated downlink subband **525-a**, the second dedicated downlink subband **525-b**, and the dedicated uplink subband **530-a**, that may span the same frequency as the uplink and downlink BWP pair. In some cases, the dedicated subbands (e.g., the first dedicated downlink subband **525-a**, the second dedicated downlink subband **525-b**, and the dedicated uplink subband **530-a**) may correspond to the set of cell-specific subbands. For example, the uplink BWP **515-a** (which may be an active uplink BWP for the UE **115**) may overlap (at least partially) with the cell uplink subband **510-a**, and the corresponding SBFD uplink subband for the UE **115** (e.g., dedicated uplink subband **530-a**) may include uplink resources located within the overlapped portion of the uplink BWP **515-a** and the cell uplink subband **510-a**. In addition, the downlink BWP **520-a** (which may be an active downlink BWP for the UE **115**) also may overlap with the cell uplink subband **510-a**, and may partially overlap with the first cell downlink subband **505-a** and the second cell downlink subband **505-b**. In such examples, the corresponding dedicated downlink subbands (e.g., first dedicated downlink subband **525-a** and second dedicated downlink subband **525-b**) may include downlink resources located within the overlapped portion of the downlink BWP **520-a**, the first cell downlink subband **505-a**, and the second cell downlink subband **505-b**.

[0127] The SBFD configuration **402** may illustrate an example BWP pair configuration including uplink BWP **515-b** and downlink BWP **520-b**, which may be configured for the UE **115** within (at least) a portion of the cell bandwidth supported by the network entity **105**, including a first cell downlink subband **505-c**, a cell uplink subband **510-b**, and a second cell downlink subband **505-d**, which may be separated from one another by a cell guard band. In some examples, the uplink BWP **515-b** and the downlink BWP **520-b** may span different bandwidths (e.g., the uplink BWP **515-b** may be relatively larger than the downlink BWP **520-b** to support increased uplink throughput, urgent traffic, or to support a relatively large quantity of uplink communications by the UE **115**).

[0128] In some examples, the UE **115** may operate within a UE-specific carrier bandwidth (which includes at least a portion of the cell bandwidth), and may receive one or more uplink and downlink SBFD configurations that configure different SBFD uplink and downlink subbands (e.g., first dedicated downlink subband **525-c**, second dedicated downlink subband **525-d**, and dedicated uplink subband **530-b**) that correspond to each uplink and downlink BWP pair within the channel bandwidth of the UE **115**. For example, the UE **115** may have dedicated subbands for SBFD, including the first dedicated downlink subband **525-c**, the second dedicated downlink subband **525-d**, and the dedicated uplink subband **530-b**, that may span the same frequency as the uplink and downlink BWP pair. In some cases, the dedicated subbands (e.g., the first dedicated downlink subband **525-c**, the second dedicated downlink subband **525-d**, and the dedicated uplink subband **530-b**) may correspond to the set of cell-specific subbands. For example, the uplink BWP **515-b** (which may be an active uplink BWP for the UE **115**) may overlap with the cell uplink subband **510-b**, and the corresponding SBFD uplink subband for the UE **115** (e.g., dedicated uplink subband **530-b**) may include uplink resources located within the overlapped portion of the uplink BWP **515-b** and the cell uplink subband **510-b**. In addition, the downlink BWP **520-b** (which may be an active

downlink BWP for the UE **115**) also may overlap with the cell uplink subband **510-b**, and may partially overlap with the first cell downlink subband **505-c** and the second cell downlink subband **505-d**. In such examples, the corresponding dedicated downlink subbands (e.g., first dedicated downlink subband **525-c** and second dedicated downlink subband **525-d**) may include downlink resources located within the overlapped portion of the downlink BWP **520-b**, the first cell downlink subband **505-c**, and the second cell downlink subband **505-d**.

[0129] In some implementations, the configuration of the dedicated uplink and downlink subbands for SBFD may be based on one or more capabilities of the UE **115**. For example, the UE **115** may support up to a threshold guard band size between the dedicated uplink subband and the dedicated downlink subbands. Additionally, or alternatively, the SBFD configuration may be based on one or more supported uplink and downlink subband sizes (e.g., the UE **115** may support one or more required sizes of the dedicated uplink subbands, the first dedicated downlink subband, and the second dedicated downlink subband). In some examples, the supported uplink and downlink subband sizes may be configured to match an existing channel bandwidth of the UE **115**, or may be configured to satisfy one or more size requirements (e.g., minimum size or maximum size). In some aspects, the UE **115** may support capability signaling in which the UE **115** may transmit one or more messages that indicate one or more capabilities of the UE **115**. For example, the UE **115** may signal a bitmap that is indicative of one or more sizes of uplink subbands, downlink subbands, or both, that the UE **115** supports.

[0130] FIG. **6** shows an example of an SBFD configuration **600** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the SBFD configuration **600** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. **1** and **2**. For example, the SBFD configuration **600** may be configured by a network entity, and may be implemented by a UE, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. **1** and **2**. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0131] In some aspects, the network entity **105** may support communications on cell-specific channel bandwidth **605**, which may include at least a portion of a frequency band of a carrier. In some examples, a relative frequency location of the cell-specific channel bandwidth **605** may be indicated with respect to a carrier offset (e.g., a cell-specific offset to carrier). The network entity **105** may perform communications using different frequencies and communications resources located at the cell-specific channel bandwidth **605**. In some cases, the cell-specific channel bandwidth **605** may be segmented into a quantity of uplink and downlink subbands. For example, the cell-specific channel bandwidth **605** may include a first downlink subband **610-a**, a second downlink subband **610-b**, and an uplink subband **615**. The network entity **105** may obtain or receive uplink communications using the uplink subband **615** (or may allocate resources for various uplink communications on the uplink subband **615**), and may perform downlink communications using the first downlink subband **610-a** and the second downlink subband **610-b** (or may allocate resources for various downlink communications on the first downlink subband **610-a** and the second downlink subband **610-b**).

[0132] In some implementations, the UE **115** may be allocated one or more active BWPs using at least a portion of the cell-specific channel bandwidth **605**. For example, the UE **115** may be allocated an active uplink BWP **620**. In such examples, the network entity **105** may configure an uplink and downlink SBFD configuration for each BWP (or BWP pair) allocated to the UE **115**. For example, the SBFD configuration may indicate an uplink subband **625** for the UE **115** to use for SBFD communications. In some other examples, the UE **115** may be allocated one or more downlink BWPs, and corresponding downlink subbands, such that the description of SBFD configuration **600** is not limited to uplink BWPs and corresponding uplink subbands.

[0133] In some aspects, the UE **115** may identify the subband frequency location (e.g., a frequency location of the uplink subband **625**) based on a resource indication value (RIV) that indicates a “start” resource block (e.g., a resource block where the subband begins) and number associated with the resource block. In some examples, the start resource block may be determined using the same subcarrier spacing of the active uplink BWP **620**. In some implementations, the start resource block of the uplink subband **625** may be determined with respect the lowest (e.g., first) physical resource block (PRB) of the carrier. For example, the UE **115** may determine (or the SBFD configuration may indicate) the common resource block where the uplink subband **625** starts (N.sub.SB.sup.start), with respect to the first resource block of the carrier, CRB0, using:

$$[00001] N_{SB}^{start} = O_{carrier} + \text{Subband} - Rb_{start},$$

where O.sub.carrier is the offset to the carrier (cell-specific) and the Subband-Rb.sub.start is the offset from the end of the offset to the carrier (cell-specific) to the beginning of the uplink subband **625**. In such examples, the RIV may be determined assuming N.sub.subband.sup.size=275 resource blocks.

[0134] In some other examples, the UE **115** may determine the start resource block of the uplink subband **625** with respect the lowest (e.g., first) PRB of the active uplink BWP **620**. The RIV may be determined based on the actual size of the active uplink BWP **620**, which may reduce overhead associated with the RIV calculation. For example, the UE **115** may determine the common resource block where the uplink subband starts (N.sub.SB.sup.start) with respect to the first resource block of the carrier, CRB0, using:

$$[00002] N_{SB}^{start} = O_{carrier} + RB - \text{start(BWP)} + \text{Subband} - Rb_{start},$$

where O.sub.carrier is the offset to the carrier (cell-specific), RB-start (BWP) is the offset from end of the offset of the carrier (cell-specific) to the beginning of the active uplink BWP **620**, and the Subband-Rb.sub.start is the offset from the beginning of the active uplink BWP **620** to the beginning of the uplink subband **625**.

[0135] In some other examples, the network entity **105** may configure the uplink and downlink SBFD configuration for each BWP pair, where each of the uplink and downlink subbands represent corresponding virtual uplink and downlink BWPs in one or more SBFD symbols. For example, the uplink and downlink subbands may include similar or identical frequency resources, size, or both, as the corresponding uplink and downlink BWPs, or may be associated with the corresponding uplink and downlink BWPs by one or more mappings. In some aspects, the association between the uplink and downlink subbands and the corresponding virtual uplink and downlink BWPs may enable efficient allocation for uplink frequency resources in terms of frequency domain resource allocation (FDRA), such as type 0/1 FDRA, intra-slot and inter-slot frequency hopping, among other frequency resource allocation techniques, since the frequency resource allocation may be the same for the uplink and downlink subbands as for one or more uplink and downlink BWPs.

[0136] In some examples, the network entity **105**, the UE **115**, or both, may determine a resource block group (RBG) size of a frequency allocation (e.g., allocation type 0, bitmap-based allocation) based on the uplink and downlink subband size in SBFD symbols (e.g., N.sub.BWP.sup.size=N.sub.subband.sup.size and). In such examples, the starting resource block of the resource allocation may be the first resource block of the uplink subband, or the downlink subband, given by N.sub.BWP.sup.start=N.sub.subband.sup.start. In such examples, the active uplink BWP **620** may be represented as a virtual BWP that has the same size and starting resource block of the uplink subband **625**.

[0137] In some other examples, the network entity **105**, the UE **115**, or both, may determine a resource block group (RBG) size of a frequency allocation (e.g., allocation type 1) using a RIV-based allocation type: In such examples, the starting resource block may correspond to a starting resource block in reference to the lowest (e.g., first) downlink or uplink subband, where N.sub.BWP.sup.size=N.sub.subband.sup.size.

[0138] In some other examples, the resource allocation may be configured with a frequency

hopping-based approach, where the starting resource block corresponds to the starting resource block in reference to the lowest (e.g., first) resource block of the uplink subband, where the resource block offset is an SBFD-specific offset or is determined based on the subband size. For example, for intra-slot frequency hopping, the starting resource block may be determined using:

$$[00003] \text{RB}_{\text{start}} = \begin{cases} \text{RB}_{\text{start}}, & i = 0 \\ (\text{RB}_{\text{start}} + \text{RB}_{\text{offset}}) \bmod N_{\text{SB}}^{\text{size}}, & i = 1 \end{cases},$$

and for inter-slot frequency hopping, the starting resource block may be determined using:

$$[00004] \text{RB}_{\text{start}} = \begin{cases} \text{RB}_{\text{start}}, & n_s = 0 \\ (\text{RB}_{\text{start}} + \text{RB}_{\text{offset}}) \bmod N_{\text{SB}}^{\text{size}}, & n_s = 1 \end{cases}.$$

[0139] FIG. 7 shows an example of a wireless communications system **700** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the wireless communications system **700** may illustrate communications between a network entity **105** and a UE **115**, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. 1 and 2. In some examples, the UE **115** and network entity **105** may support SBFD communications including uplink and downlink signaling in an allocated frequency band.

[0140] In aspects, the network entity **105** may signal or otherwise indicate an uplink and downlink SBFD configuration as a single cell-common configuration, or using a per-BWP configuration. For example, to support efficient SBFD resource allocation and communication, the network entity **105** may configure an uplink and downlink SBFD configuration for each uplink and downlink BWP pair (including an uplink BWP and a downlink BWP) configured for the UE **115**. In some implementations, the uplink and downlink SBFD configuration may be indicative of a resource allocation associated with SBFD communications for the UE **115**. For example, each uplink BWP may correspond to a dedicated uplink subband for SBFD communications, and each downlink BWP may correspond to a dedicated downlink subband for SBFD communications.

[0141] In some implementations, the network entity **105** may configure SBFD-dedicated uplink and downlink signals, and the associated channels that support the UE-dedicated uplink and downlink subbands. In some examples, the SBFD configuration may be included in a first information element **705** for the uplink BWP (e.g., BWP-Uplink), in a second information element **710** for the downlink BWP (e.g., BWP-Downlink), or both. For example, each of the BWP-Uplink and BWP-Downlink may include an SBFD configuration field and a SBFD configuration identifier (e.g., SBFD-Configuration, SBFD-Configuration-id) that may configure the dedicated uplink and downlink subbands for the uplink and downlink BWPs. Additionally, or alternatively, the SBFD configuration may be different uplink and downlink SBFD configurations such that each of the BWP-Uplink and BWP-Downlink may include a corresponding uplink or downlink SBFD configuration field and a SBFD configuration identifier (e.g., UL-SBFD-Configuration, DL-SBFD-Configuration, SBFD-Configuration-id).

[0142] For example, a BWP-Uplink information element may include:

TABLE-US-00001 BWP-Uplink ::= SEQUENCE { bwp-Id BWP-Id, bwp-Common BWP-UplinkCommon bwp-Dedicated BWP-UplinkDedicated SBFD-Configuration SBFD-configuration_id ... },

where the SBFD-Configuration and the SBFD-configuration_id indicate the SBFD configuration for one or more dedicated uplink subbands. Additionally, or alternatively, a downlink a BWP-Downlink information element may include:

TABLE-US-00002 BWP-Downlink ::= SEQUENCE { bwp-Id BWP-Id, bwp-Common BWP-DownlinkCommon bwp-Dedicated BWP-DownlinkDedicated SBFD-Configuration SBFD-configuration_id ... }

where the SBFD-Configuration and the SBFD-configuration_id indicate the SBFD configuration

for one or more dedicated downlink subbands.

[0143] The SBFD configuration may include various parameters, including uplink and downlink subband locations, subcarrier spacing information, time locations, semi-persistent scheduling configurations, uplink channel configurations, configured grant configurations, sounding reference signal configurations, among other parameters. For example, an SBFD-Configuration field may include:

```
TABLE-US-00003 SBFD-configuration ::= SEQUENCE { SBFD-configuration_id  
SBs-Id, UL-subband-locationAndBW (RIV) DL-subband_1-locationAndBW (RIV) DL-  
subband_2-locationAndBW (RIV) Subband-subcarrierSpacing SBFD_time_locations sps-  
Config SetupRelease { SPS-Config } pucch-Config SetupRelease { PUCCH-Config }  
pusch-Config SetupRelease { PUSCH-Config } configuredGrantConfig SetupRelease  
{ ConfiguredGrantConfig } srs-Config SetupRelease { SRS-Config } }
```

[0144] In some other examples, a BWP-Uplink information element may include a specific uplink SBFD configuration (UL-SBFD-Configuration), and may include:

```
TABLE-US-00004 BWP-Uplink ::= SEQUENCE { bwp-Id BWP-Id, bwp-Common  
BWP-UplinkCommon bwp-Dedicated BWP-UplinkDedicated UL-SBFD-Configuration  
SBFD-configuration_id ... },
```

where the UL-SBFD-Configuration and the SBFD-configuration_id indicate a dedicated uplink SBFD configuration for one or more dedicated uplink subbands. In such examples, the UL-SBFD-Configuration may include various parameters, including uplink subband locations, subcarrier spacing information, time locations, semi-persistent scheduling configurations, uplink channel configurations, configured grant configurations, sounding reference signal configurations, among other parameters. For example, an UL-SBFD-Configuration field may include:

```
TABLE-US-00005 UL-SBFD-configuration ::= SEQUENCE { UL-SBFD-configuration_id  
SBs-Id, UL-subband-locationAndBW (RIV) Subband-subcarrierSpacing  
SBFD_time_locations pucch-Config SetupRelease { PUCCH-Config } pusch-Config  
SetupRelease { PUSCH-Config } configuredGrantConfig SetupRelease {  
ConfiguredGrantConfig } srs-Config SetupRelease { SRS-Config } }.
```

Additionally, or alternatively, a downlink a BWP-Downlink information element may include a specific downlink SBFD configuration (DL-SBFD-Configuration), and may include:

```
TABLE-US-00006 BWP-Downlink ::= SEQUENCE { bwp-Id BWP-Id, bwp-Common  
BWP-DownlinkCommon bwp-Dedicated BWP-DownlinkDedicated DL-SBFD-Configuration  
SBFD-configuration_id ... }
```

where the DL-SBFD-Configuration and the SBFD-configuration_id indicate a dedicated downlink SBFD configuration for one or more dedicated downlink subbands. In such examples, the DL-SBFD-Configuration may include various parameters, including downlink subband locations, subcarrier spacing information, time locations, semi-persistent scheduling configurations, downlink channel configurations, configured grant configurations, sounding reference signal configurations, among other parameters. For example, a DL-SBFD-Configuration field may include:

```
TABLE-US-00007 DL-SBFD-configuration ::= SEQUENCE { SBFD-configuration_id SBs-Id,  
DL-subband_1-locationAndBW (RIV) DL-subband_2-locationAndBW (RIV) Subband-  
subcarrierSpacing sps-Config SetupRelease { SPS-Config } pdcch-Config SetupRelease {  
PDCCH-Config } pdsch-Config SetupRelease { PDSCH-Config } }.
```

[0145] FIG. 8 shows an example of a SBFD configuration **801**, a SBFD configuration **802**, and a SBFD configuration **803** that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the SBFD configuration **801**, SBFD configuration **802**, and SBFD configuration **803** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. 1 and 2. For example, the SBFD configuration **801**, SBFD configuration **802**, and SBFD configuration **803** may be configured by a network entity, and may be implemented by a

UE, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. **1** and **2**. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0146] In some implementations, a network entity may configure a cell-common uplink and downlink SBFD configuration, where resources for the uplink and downlink subbands for SBFD communications (e.g., frequency locations of the uplink and downlink subbands) may be defined within a common resource block grid of a cell. In SBFD configuration **801**, for example, a network entity may support communications on cell-specific channel bandwidth **805-a**, which may include at least a portion of a frequency band of a carrier. In some examples, a relative frequency location of the cell-specific channel bandwidth **805-a** may be indicated with respect to a carrier offset (e.g., a cell-specific offset to carrier). The network entity may perform communications using different frequencies and communications resources located at the cell-specific channel bandwidth **805-a**. In some cases, the cell-specific channel bandwidth **805-a** may be segmented into a quantity of uplink and downlink subbands. For example, the cell-specific channel bandwidth **805-a** may include a first downlink subband **810-a**, a second downlink subband **810-b**, and an uplink subband **815-a**. The network entity may obtain uplink communications using the uplink subband **815-a** (or may allocate resources for various uplink communications on the uplink subband **815-a**), and may output downlink communications using the first downlink subband **810-a** and the second downlink subband **810-b** (or may allocate resources for various downlink communications on the first downlink subband **810-a** and the second downlink subband **810-b**).

[0147] In some implementations, a UE may be allocated a UE channel specific bandwidth **820-a**, which may include at least a portion of the cell-specific channel bandwidth **805-a**. In such examples, the network entity may configure one or more cell-specific uplink and downlink SBFD configurations, where the SBFD configuration may be specific to the cell-specific channel bandwidth **805-a**, or the UE channel specific bandwidth **820-a**. For example, the SBFD configuration may indicate a first dedicated downlink subband **825-a**, a second dedicated downlink subband **825-b**, and a dedicated uplink subband **830-a** for the UE **115** to use for SBFD communications. In cases where the SBFD configuration is associated with the cell-specific channel bandwidth **805-a**, the dedicated uplink and downlink subbands may be indicated within the common resource blocks of the cell-specific channel bandwidth (e.g., the cell-specific channel bandwidth **805-a**). Additionally, or alternatively, in cases where the SBFD configuration is associated with the UE channel specific bandwidth **820-a**, the dedicated uplink and downlink subbands may be provided within the common resource blocks of the UE-dedicated channel bandwidth (e.g., the UE channel specific bandwidth **820-a**).

[0148] In SBFD configuration **802**, the network entity may support communications on cell-specific channel bandwidth **805-b**, which may include at least a portion of a frequency band of a carrier. In some examples, a relative frequency location of the cell-specific channel bandwidth **805-b** may be indicated with respect to a carrier offset (e.g., a cell-specific offset to carrier). The network entity may perform communications using different frequencies and communications resources located at the cell-specific channel bandwidth **805-b**. In some cases, the cell-specific channel bandwidth **805-b** may be segmented into a quantity of uplink and downlink subbands. For example, the cell-specific channel bandwidth **805-b** may include a first downlink subband **810-c**, a second downlink subband **810-d**, and an uplink subband **815-b**. The network entity may obtain uplink communications using the uplink subband **815-b** (or may allocate resources for various uplink communications on the uplink subband **815-b**), and may perform or output downlink communications using the first downlink subband **810-c** and the second downlink subband **810-d** (or may allocate resources for various downlink communications on the first downlink subband **810-c** and the second downlink subband **810-d**).

[0149] In some implementations, a UE may be allocated one or more active uplink and downlink

BWPs (e.g., active downlink BWP **835-a** and active uplink BWP **840-a**), which may be allocated to the UE using at least a portion of the cell-specific channel bandwidth **805-b**. In such examples, the network entity may configure one or more cell-common uplink and downlink SBF configurations, where the UE may implicitly determine one or more usable or available resources for uplink and downlink communications (e.g., SBF communications) per the active downlink BWP **835-a** and active uplink BWP **840-a**, respectively. In some examples, the UE may determine the SBF configuration for uplink and downlink communications based on the overlapping frequency resources (e.g., PRBs) between the downlink and uplink SBs (e.g., first dedicated downlink subband **825-c**, second dedicated downlink subband **825-d**, and dedicated uplink subband **830-b**) and the active downlink and uplink BWPs. In some examples, The UE may receive communications in downlink frequency resources of the first dedicated downlink subband **825-c**, the second dedicated downlink subband **825-d**, or both, which may be within the active downlink BWP **835-a**. Additionally, or alternatively, the UE may transmit communications in uplink frequency resources of the dedicated uplink subband **830-b** which may be within the active uplink BWP **840-a**.

[0150] In SBF configuration **803**, the network entity may support communications on cell-specific channel bandwidth **805-c**, which may include at least a portion of a frequency band of a carrier. In some examples, a relative frequency location of the cell-specific channel bandwidth **805-c** may be indicated with respect to a carrier offset (e.g., a cell-specific offset to carrier). The network entity may perform communications using different frequencies and communications resources located at the cell-specific channel bandwidth **805-c**. In some cases, the cell-specific channel bandwidth **805-c** may be segmented into a UE channel specific bandwidth **820-b**, which may be further segmented into a quantity of uplink and downlink subbands. For example, the UE channel specific bandwidth **820-b** may include a first downlink subband **810-e**, a second downlink subband **810-f**, and an uplink subband **815-c**. The UE may perform uplink communications using the uplink subband **815-c**, and may receive downlink communications using the first downlink subband **810-e** and the second downlink subband **810-f**.

[0151] In some implementations, a UE may be allocated one or more active uplink and downlink BWPs (e.g., active downlink BWP **835-b** and active uplink BWP **840-b**), which may be allocated to the UE using at least a portion of the UE channel specific bandwidth **820-b**. In such examples, the network entity may configure one or more cell-common uplink and downlink SBF configurations, where the UE may implicitly determine one or more usable or available resources for uplink and downlink communications (e.g., SBF communications) per the active downlink BWP **835-b** and active uplink BWP **840-b**, respectively. In some examples, the UE may determine the SBF configuration for uplink and downlink communications based on the overlapping frequency resources (e.g., PRBs) between the downlink and uplink subbands (e.g., first dedicated downlink subband **825-e**, second dedicated downlink subband **825-f**, and dedicated uplink subband **830-c**) and the active downlink and uplink BWPs. In some examples, The UE may receive communications in downlink frequency resources of the first dedicated downlink subband **825-e**, the second dedicated downlink subband **825-f**, or both, which may be within the active downlink BWP **835-b**. Additionally, or alternatively, the UE may transmit communications in uplink frequency resources of the dedicated uplink subband **830-c** which may be within the active uplink BWP **840-b**.

[0152] In some implementations, a network entity may indicate one or more SBF time patterns or time resources for a UE that is in a connected (e.g., RRC connected) state. For example, the network entity may indicate the one or more SBF time patterns or time resources in a UE-dedicated cell common configuration (e.g., via a ServingCellConfig information element). For example, the UE-dedicated cell common configuration may be used to configure (e.g., add or modify) the UE with a serving cell, and may indicate a set of cell-specific parameters including the SBF time patterns or resources. In some aspects, the UE-dedicated cell common configuration

may include aspects of a downlink-uplink time division duplexing configuration, where the uplink and downlink subbands may have a subcarrier spacing that is different from a subcarrier spacing of a corresponding BWP.

[0153] In some other examples, the network entity may indicate the one or more SBFD time patterns or time resources in a UE-dedicated configuration per each uplink and downlink BWP configuration using the same BWP subcarrier spacing. In some other examples, the network entity may indicate the one or more SBFD time patterns or time resources in a cell common configuration broadcasted via a system information message (e.g., a SIB message). For example, the SBFD time patterns or time resources may be indicated via a common serving cell configuration (e.g., via the ServingCellConfigCommon information element), with subcarrier spacing that is the same as the subcarrier spacing associated with TDD-DL-UL common. In such examples, the UE may be in an active state, an idle state, or an inactive state. Additionally, or alternatively, the SBFD pattern periodicity may be calculated or otherwise determined based on the TDD-DL-UL periodicity, or the SBFD pattern periodicity may be explicitly indicated or configured by the network entity.

[0154] FIG. 9 shows an example of a SBFD configuration **901** and SBFD configuration **902** that support subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. For example, the SBFD configuration **901** and SBFD configuration **902** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. 1 and 2. For example, the SBFD configuration **901** and SBFD configuration **902** may be configured by a network entity, and may be implemented by a UE, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. 1 and 2. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0155] In some implementations, a network entity may configure SBFD configurations per allocated BWP of a UE. For example, the UE may use a set of designated uplink subbands, downlink subbands, or both, that correspond to resources within an active uplink or downlink BWP configured for the UE. In some cases, however, a subcarrier spacing of the downlink BWP, the uplink BWP, or both, may be different from the subcarrier spacing of the dedicated uplink subband, the dedicated downlink subband, or both, as illustrated in SBFD configuration **901**. For example, a set of uplink or downlink resource blocks **905-a** of an uplink or downlink subband may be configured for uplink or downlink communications. In addition, a corresponding set of uplink or downlink resource blocks **910-a** of a corresponding uplink or downlink BWP may also be configured. In some aspects, one or more resource blocks of the uplink or downlink BWP may extend past the allocated resource blocks of the uplink or downlink subband. In such examples where the subcarrier spacing of the uplink or downlink BWP is greater than the subcarrier spacing of the designated uplink or downlink, the edge resource blocks may be dropped (e.g., dropped RB **915-a** and dropped RB **915-b**).

[0156] In some other cases, a subcarrier spacing of the downlink BWP, the uplink BWP, or both, may be smaller than the subcarrier spacing of the dedicated uplink subband, the dedicated downlink subband, or both, as illustrated in SBFD configuration **902**. For example, a set of uplink or downlink resource blocks **905-b** of an uplink or downlink subband may be configured for uplink or downlink communications. In addition, a corresponding set of uplink or downlink resource blocks **910-b** of a corresponding uplink or downlink BWP may also be configured. In some aspects, the resource blocks of the uplink or downlink BWP may be included in the allocated resource blocks of the uplink or downlink subband. In such examples where the subcarrier spacing of the uplink or downlink BWP is smaller than the subcarrier spacing of the designated uplink or downlink, the UE may refrain from dropping resource blocks.

[0157] FIG. 10 shows an example of a SBFD configuration **1000** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present

disclosure. For example, the SBFD configuration **1000** may be implemented at or by aspects of wireless communications system **100** and wireless communications system **200**, described with reference to FIGS. **1** and **2**. For example, the SBFD configuration **1000** may be configured by a network entity, and may be implemented by a UE, each of which may be examples of corresponding network entities **105** and UEs **115** as described with reference to FIGS. **1** and **2**. In some examples, the UE **115** and network entity **105** may support SBFD communications including simultaneous uplink and downlink signaling in an allocated frequency band.

[0158] In some aspects, the network entity **105** may support communications on cell-specific channel bandwidth **1005**, which may include at least a portion of a frequency band of a carrier. In some examples, a relative frequency location of the cell-specific channel bandwidth **1005** may be indicated with respect to a carrier offset (e.g., a cell-specific offset to carrier). The network entity **105** may perform communications using different frequencies and communications resources located at the cell-specific channel bandwidth **1005**. In some cases, the cell-specific channel bandwidth **1005** may be segmented into a quantity of uplink and downlink subbands. For example, the cell-specific channel bandwidth **1005** may include a first downlink subband **1010-a**, a second downlink subband **1010-b**, and an uplink subband **1015**. The network entity **105** may obtain uplink communications using the uplink subband **1015** (or may allocate resources for various uplink communications on the uplink subband **1015**), and may perform or output downlink communications using the first downlink subband **1010-a** and the second downlink subband **1010-b** (or may allocate resources for various downlink communications on the first downlink subband **1010-a** and the second downlink subband **1010-b**).

[0159] In some implementations, the UE **115** may be allocated one or more active BWPs using at least a portion of the cell-specific channel bandwidth **1005**. For example, the UE **115** may be allocated an active downlink BWP **1020** and an active uplink BWP **1025**. In such examples, the network entity **105** may configure an uplink and downlink SBFD configuration for each BWP (or BWP pair) allocated to the UE **115**. In some examples, the UE channel bandwidth may be configured for the UE **115** after RRC connection, where during an initial access procedure the UE **115** identifies the serving cell channel bandwidth and location, in addition to one or more initial BWP locations. In some cases, however, the UE **115** may be operating in an idle or inactive state, and the network entity **105** may use broadcast signaling (among other techniques) to configure the UE **115** with uplink and downlink subband configurations. For example, the network entity **105** may transmit one or more system information blocks (e.g., SIB1) that includes the time and frequency configurations of the dedicated uplink and downlink subbands. In some examples, the uplink and downlink subband locations may be configured or defined per the initial uplink and downlink BWP cell-specific channel bandwidth (e.g., the downlink and uplink subbands may be included within the common resource blocks of the cell-specific channel bandwidth). Additionally, or alternatively, configurations for the initial downlink subband **1030** and the initial uplink subband **1035** may be indicated per the serving cell bandwidth (e.g., within the resource blocks of the initial channel bandwidth).

[0160] FIG. **11** shows an example of a process flow **1100** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The process flow **1100** may implement or be implemented by aspects of the wireless communications system **100** and the wireless communications system **200**. The process flow **1100** may include a network entity **105** and a UE **115**, each of which may be examples of network entities and UEs described herein. In the following description of process flow **1100**, the operations may be performed in a different order than the order shown, or other operations may be added or removed from the process flow **1100**. For example, some operations may also be left out of process flow **1100**, may be performed in different orders or at different times, or other operations may be added to process flow **1100**. Although communications of the process flow **1100** are shown occurring between a network entity **105** and a UE **115**, some aspects of some operations may also be performed by one

or more other wireless devices, network devices, or network functions.

[0161] At **1105**, the UE **115** may receive first information that is indicative of one or more BWP pairs configured for the UE **115**, where each BWP pair of the one or more BWP pairs includes a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE **115**.

[0162] At **1110**, the UE **115** may receive second information that is indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs, which may include one or more active BWPs. In some examples, the UE **115** may transmit one or more UE capabilities via capability signaling prior to receiving the second information. In such examples, the UE **115** may receive the second information indicative of the SBFD configuration, where the SBFD configuration is based on (or configured in accordance with) the one or more UE capabilities. In some aspects, the one or more capabilities may include (but may not be limited to) a supported guard band size between the uplink subband and the downlink subband, a threshold size of the uplink subband supported by the UE **115**, a threshold size of the downlink subband supported by the UE **115**, or any combination thereof.

[0163] In some implementations, the uplink subband may correspond to a respective virtual uplink BWP and the downlink subband may correspond to a respective virtual downlink BWP, and the UE **115** may receive the second information indicative of the SBFD configuration by receiving an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, where a resource block group size of the resource allocation is based on a size of the uplink subband, a size of the downlink subband, or both.

[0164] In some implementations, the uplink subband may correspond to a respective virtual uplink BWP and the downlink subband may correspond to a respective virtual downlink BWP, and receiving the second information indicative of the SBFD configuration includes receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both. In such implementations, the resource allocation of the scheduled physical channel may include a RIV that includes an indication of a starting resource block of the resource allocation that is with reference to a first resource block of the uplink subband or a first resource block of the downlink subband. In some other implementations, the UE **115**, as part of the second information, may receive an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, and the physical channel resource allocation indicates a starting resource block of each hop of the frequency hopping relative to a first resource block of the uplink subband. In addition, a resource block offset may be based on a size of the uplink subband, the downlink subband, or both.

[0165] In some aspects, receiving the second information indicative of the SBFD configuration may include receiving respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, where the SBFD configuration is indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration. In some examples, the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more RIVs, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

[0166] In some implementations, the SBFD configuration may be configured in accordance with a common resource block grid of a cell (e.g., a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both) configured for SBFD communications.

[0167] In some implementations, the UE **115** may be operating in a connected state (e.g., an RRC connected state), and the UE **115** may receive an SBFD configuration that includes a time pattern

or an indication of resources for SBFD communications. In some examples, the time pattern or indication of resources may be for each of the one or more BWP pairs. In some examples, the time pattern or indication of the resources may be received via a cell common configuration of a broadcast message such as a SIB. In some other examples, the UE **115** may receive broadcast signaling that indicates that the uplink subband, the downlink subband, or both, are configured relative to an initial BWP, and are configured based at least in part on a serving cell bandwidth, or both.

[0168] At **1115**, the UE **115** may communicate one or more messages via the one or more BWP pairs in accordance with the SBFD configuration. In some examples, the SBFD configuration may include a RIV that indicates a resource allocation associated with the uplink subband, the downlink subband or both, and the RIV may also include an indication of a starting resource block of the resource allocation (e.g., indicated relative to a first physical resource block of a carrier bandwidth, or relative to a first physical resource block of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE **115**) and a quantity of resource blocks included in the resource allocation.

[0169] In some aspects, the UE **115** may communicate the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based on one or more physical resource blocks that at least partially overlap with the respective uplink BWP, the respective downlink BWP, or both, of the SBFD configuration. In some other examples, the UE **115** may drop one or more physical resource blocks of the respective uplink BWP, the respective downlink BWP, or both, based on a subcarrier spacing of the respective uplink BWP, the respective downlink BWP, or both, being greater than a corresponding subcarrier spacing of the uplink subband, the downlink subband, or both.

[0170] FIG. **12** shows a block diagram **1200** of a device **1205** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The device **1205** may be an example of aspects of a UE **115** as described herein. The device **1205** may include a receiver **1210**, a transmitter **1215**, and a communications manager **1220**. The device **1205**, or one or more components of the device **1205** (e.g., the receiver **1210**, the transmitter **1215**, the communications manager **1220**), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0171] The receiver **1210** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to subband configuration techniques for SBFD). Information may be passed on to other components of the device **1205**. The receiver **1210** may utilize a single antenna or a set of multiple antennas.

[0172] The transmitter **1215** may provide a means for transmitting signals generated by other components of the device **1205**. For example, the transmitter **1215** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to subband configuration techniques for SBFD). In some examples, the transmitter **1215** may be co-located with a receiver **1210** in a transceiver module. The transmitter **1215** may utilize a single antenna or a set of multiple antennas.

[0173] The communications manager **1220**, the receiver **1210**, the transmitter **1215**, or various combinations or components thereof may be examples of means for performing various aspects of subband configuration techniques for SBFD as described herein. For example, the communications manager **1220**, the receiver **1210**, the transmitter **1215**, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0174] In some examples, the communications manager **1220**, the receiver **1210**, the transmitter **1215**, or various combinations or components thereof may be implemented in hardware (e.g., in

communications management circuitry). The hardware may include at least one of a processor, a digital signal processor (DSP), a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some examples, at least one processor and at least one memory coupled with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0175] Additionally, or alternatively, the communications manager **1220**, the receiver **1210**, the transmitter **1215**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager **1220**, the receiver **1210**, the transmitter **1215**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0176] In some examples, the communications manager **1220** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1210**, the transmitter **1215**, or both. For example, the communications manager **1220** may receive information from the receiver **1210**, send information to the transmitter **1215**, or be integrated in combination with the receiver **1210**, the transmitter **1215**, or both to obtain information, output information, or perform various other operations as described herein.

[0177] The communications manager **1220** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1220** is capable of, configured to, or operable to support a means for receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The communications manager **1220** is capable of, configured to, or operable to support a means for receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The communications manager **1220** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0178] By including or configuring the communications manager **1220** in accordance with examples as described herein, the device **1205** (e.g., at least one processor controlling or otherwise coupled with the receiver **1210**, the transmitter **1215**, the communications manager **1220**, or a combination thereof) may support techniques for reduced processing, reduced ambiguity for resource allocation for SBFD communications, and more efficient utilization of uplink and downlink communication resources based on per-BWP configuration of SBFD communications.

[0179] FIG. **13** shows a block diagram **1300** of a device **1305** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The device **1305** may be an example of aspects of a device **1205** or a UE **115** as described herein. The device **1305** may include a receiver **1310**, a transmitter **1315**, and a communications manager **1320**. The device **1305**, or one or more components of the device **1305** (e.g., the receiver **1310**, the transmitter **1315**, the communications manager **1320**), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components

may be in communication with one another (e.g., via one or more buses).

[0180] The receiver **1310** may provide a means for receiving information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to subband configuration techniques for SBFD). Information may be passed on to other components of the device **1305**. The receiver **1310** may utilize a single antenna or a set of multiple antennas.

[0181] The transmitter **1315** may provide a means for transmitting signals generated by other components of the device **1305**. For example, the transmitter **1315** may transmit information such as packets, user data, control information, or any combination thereof associated with various information channels (e.g., control channels, data channels, information channels related to subband configuration techniques for SBFD). In some examples, the transmitter **1315** may be co-located with a receiver **1310** in a transceiver module. The transmitter **1315** may utilize a single antenna or a set of multiple antennas.

[0182] The device **1305**, or various components thereof, may be an example of means for performing various aspects of subband configuration techniques for SBFD as described herein. For example, the communications manager **1320** may include a BWP configuration component **1325**, an SBFD subband configuration component **1330**, an SBFD communication component **1335**, or any combination thereof. The communications manager **1320** may be an example of aspects of a communications manager **1220** as described herein. In some examples, the communications manager **1320**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1310**, the transmitter **1315**, or both. For example, the communications manager **1320** may receive information from the receiver **1310**, send information to the transmitter **1315**, or be integrated in combination with the receiver **1310**, the transmitter **1315**, or both to obtain information, output information, or perform various other operations as described herein.

[0183] The communications manager **1320** may support wireless communications in accordance with examples as disclosed herein. The BWP configuration component **1325** is capable of, configured to, or operable to support a means for receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The SBFD subband configuration component **1330** is capable of, configured to, or operable to support a means for receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The SBFD communication component **1335** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0184] FIG. **14** shows a block diagram **1400** of a communications manager **1420** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The communications manager **1420** may be an example of aspects of a communications manager **1220**, a communications manager **1320**, or both, as described herein. The communications manager **1420**, or various components thereof, may be an example of means for performing various aspects of subband configuration techniques for SBFD as described herein. For example, the communications manager **1420** may include a BWP configuration component **1425**, an SBFD subband configuration component **1430**, an SBFD communication component **1435**, a UE capability signaling component **1440**, a resource allocation management component **1445**, an SCS evaluation component **1450**, or any combination thereof. Each of these components, or components or subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0185] The communications manager **1420** may support wireless communications in accordance

with examples as disclosed herein. The BWP configuration component **1425** is capable of, configured to, or operable to support a means for receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The SBFD communication component **1435** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0186] In some examples, to support receiving the second information, the UE capability signaling component **1440** is capable of, configured to, or operable to support a means for transmitting an indication of one or more capabilities of the UE. In some examples, to support receiving the second information, the SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving, responsive to the indication, the second information indicative of the SBFD configuration, where the SBFD configuration is based on the one or more capabilities of the UE.

[0187] In some examples, the SBFD configuration includes a resource indication value that indicates a resource allocation associated with the uplink subband, the downlink subband or both. In some examples, the resource indication value includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation. In some examples, the starting resource block is indicated relative to a first physical resource block of a carrier bandwidth.

[0188] In some examples, the starting resource block is indicated relative to a first physical resource block of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE. In some examples, the uplink subband, the downlink subband, or both, are configured in an active BWP pair of the one or more BWP pairs.

[0189] In some examples, the SBFD configuration is based on one or more capabilities of the UE, the one or more capabilities including a supported guard band size between the uplink subband and the downlink subband, a threshold size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.

[0190] In some examples, to support receiving the second information indicative of the SBFD configuration, the resource allocation management component **1445** is capable of, configured to, or operable to support a means for receiving an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, where a resource block group size of the resource allocation is based on a size of the uplink subband, a size of the downlink subband, or both.

[0191] In some examples, to support receiving the second information indicative of the SBFD configuration, the resource allocation management component **1445** is capable of, configured to, or operable to support a means for receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, where the resource allocation of the scheduled physical channel includes a resource indication value that includes an indication of a starting resource block of the resource allocation that is with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.

[0192] In some examples, to support receiving the second information indicative of the SBFD configuration, the resource allocation management component **1445** is capable of, configured to, or operable to support a means for receiving an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, where the physical channel resource allocation indicates a starting resource block of each hop of

the frequency hopping relative to a first resource block of the uplink subband, and where a resource block offset is based on a size of the uplink subband, the downlink subband, or both.

[0193] In some examples, to support receiving the second information indicative of the SBFD configuration, the SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving the second information indicative of the SBFD configuration in respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, where the SBFD configuration is indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration.

[0194] In some examples, the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more resource indication values, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

[0195] In some examples, the SBFD configuration including the uplink subband, the downlink subband or both, is configured in accordance with a common resource block grid of a cell configured for SBFD communications. In some examples, the cell includes a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

[0196] In some examples, to support communicating the one or more messages, the SBFD communication component **1435** is capable of, configured to, or operable to support a means for communicating the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based on one or more physical resource blocks that at least partially overlap with the respective uplink BWP, the respective downlink BWP, or both, of the SBFD configuration.

[0197] In some examples, the SCS evaluation component **1450** is capable of, configured to, or operable to support a means for dropping one or more physical resource blocks of the respective uplink BWP, the respective downlink BWP, or both, based on a subcarrier spacing of the respective uplink BWP, the respective downlink BWP, or both, being greater than a corresponding subcarrier spacing of the uplink subband, the downlink subband, or both.

[0198] In some examples, to support receiving the second information indicative of the SBFD configuration, the SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving, while the UE is in a connected state, the second information indicative of the SBFD configuration via a serving cell configuration message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0199] In some examples, to support receiving the second information indicative of the SBFD configuration, the SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving, while the UE is in a connected state, the second information indicative of the SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0200] In some examples, to support receiving the second information indicative of the SBFD configuration, the SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving, while the UE is in a connected state, an idle state, or an inactive state, the second information indicative of the SBFD configuration via a cell common configuration of a broadcast message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications. In some examples, receiving the SBFD configuration includes a SBFD pattern periodicity that is based on a time division duplexing (TDD) downlink and uplink periodicity.

[0201] In some examples, the UE is operating in an inactive state or an idle state and, to support receiving the second information indicative of the SBFD configuration, the SBFD subband configuration component **1430** is capable of, configured to, or operable to support a means for receiving the second information indicative of the SBFD configuration via one or more broadcast system information messages, where the uplink subband, the downlink subband, or both, are configured relative to an initial BWP, are configured based on a serving cell bandwidth, or both.

[0202] FIG. **15** shows a diagram of a system **1500** including a device **1505** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The device **1505** may be an example of or include components of a device **1205**, a device **1305**, or a UE **115** as described herein. The device **1505** may communicate (e.g., wirelessly) with one or more other devices (e.g., network entities **105**, UEs **115**, or a combination thereof). The device **1505** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communications manager **1520**, an input/output (I/O) controller, such as an I/O controller **1510**, a transceiver **1515**, one or more antennas **1525**, at least one memory **1530**, code **1535**, and at least one processor **1540**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **1545**).

[0203] The I/O controller **1510** may manage input and output signals for the device **1505**. The I/O controller **1510** may also manage peripherals not integrated into the device **1505**. In some cases, the I/O controller **1510** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **1510** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **1510** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **1510** may be implemented as part of one or more processors, such as the at least one processor **1540**. In some cases, a user may interact with the device **1505** via the I/O controller **1510** or via hardware components controlled by the I/O controller **1510**.

[0204] In some cases, the device **1505** may include a single antenna. However, in some other cases, the device **1505** may have more than one antenna, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **1515** may communicate bi-directionally via the one or more antennas **1525** using wired or wireless links as described herein. For example, the transceiver **1515** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1515** may also include a modem to modulate the packets, to provide the modulated packets to one or more antennas **1525** for transmission, and to demodulate packets received from the one or more antennas **1525**. The transceiver **1515**, or the transceiver **1515** and one or more antennas **1525**, may be an example of a transmitter **1215**, a transmitter **1315**, a receiver **1210**, a receiver **1310**, or any combination thereof or component thereof, as described herein.

[0205] The at least one memory **1530** may include random access memory (RAM) and read-only memory (ROM). The at least one memory **1530** may store computer-readable, computer-executable, or processor-executable code, such as the code **1535**. The code **1535** may include instructions that, when executed by the at least one processor **1540**, cause the device **1505** to perform various functions described herein. The code **1535** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code **1535** may not be directly executable by the at least one processor **1540** but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory **1530** may include, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

[0206] The at least one processor **1540** may include one or more intelligent hardware devices (e.g.,

one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPU) (also referred to as neural network processors or deep learning processors (DLPs)), one or more microcontrollers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor **1540** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the at least one processor **1540**. The at least one processor **1540** may be configured to execute computer-readable instructions stored in a memory (e.g., the at least one memory **1530**) to cause the device **1505** to perform various functions (e.g., functions or tasks supporting subband configuration techniques for SBFD). For example, the device **1505** or a component of the device **1505** may include at least one processor **1540** and at least one memory **1530** coupled with or to the at least one processor **1540**, the at least one processor **1540** and the at least one memory **1530** configured to perform various functions described herein. In some examples, the at least one processor **1540** may include multiple processors and the at least one memory **1530** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions described herein. In some examples, the at least one processor **1540** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **1540**) and memory circuitry (which may include the at least one memory **1530**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **1540** or a processing system including the at least one processor **1540** may be configured to, configurable to, or operable to cause the device **1505** to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code **1535** (e.g., processor-executable code) stored in the at least one memory **1530** or otherwise, to perform one or more of the functions described herein.

[0207] The communications manager **1520** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1520** is capable of, configured to, or operable to support a means for receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The communications manager **1520** is capable of, configured to, or operable to support a means for receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The communications manager **1520** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0208] By including or configuring the communications manager **1520** in accordance with examples as described herein, the device **1505** may support techniques for improved communication reliability, reduced latency due to enhanced SBFD communications techniques and resource allocation, more efficient utilization of uplink and downlink communication resources in SBFD, improved coordination between devices, reduced ambiguity of resource allocation for SBFD uplink and downlink subbands, among other advantages.

[0209] In some examples, the communications manager **1520** may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with

the transceiver **1515**, the one or more antennas **1525**, or any combination thereof. Although the communications manager **1520** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1520** may be supported by or performed by the at least one processor **1540**, the at least one memory **1530**, the code **1535**, or any combination thereof. For example, the code **1535** may include instructions executable by the at least one processor **1540** to cause the device **1505** to perform various aspects of subband configuration techniques for SBFD as described herein, or the at least one processor **1540** and the at least one memory **1530** may be otherwise configured to, individually or collectively, perform or support such operations.

[0210] FIG. **16** shows a block diagram **1600** of a device **1605** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The device **1605** may be an example of aspects of a network entity **105** as described herein. The device **1605** may include a receiver **1610**, a transmitter **1615**, and a communications manager **1620**. The device **1605**, or one or more components of the device **1605** (e.g., the receiver **1610**, the transmitter **1615**, the communications manager **1620**), may include at least one processor, which may be coupled with at least one memory, to, individually or collectively, support or enable the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0211] The receiver **1610** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **1605**. In some examples, the receiver **1610** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **1610** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0212] The transmitter **1615** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **1605**. For example, the transmitter **1615** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **1615** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **1615** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **1615** and the receiver **1610** may be co-located in a transceiver, which may include or be coupled with a modem.

[0213] The communications manager **1620**, the receiver **1610**, the transmitter **1615**, or various combinations or components thereof may be examples of means for performing various aspects of subband configuration techniques for SBFD as described herein. For example, the communications manager **1620**, the receiver **1610**, the transmitter **1615**, or various combinations or components thereof may be capable of performing one or more of the functions described herein.

[0214] In some examples, the communications manager **1620**, the receiver **1610**, the transmitter **1615**, or various combinations or components thereof may be implemented in hardware (e.g., in communications management circuitry). The hardware may include at least one of a processor, a DSP, a CPU, an ASIC, an FPGA or other programmable logic device, a microcontroller, discrete gate or transistor logic, discrete hardware components, or any combination thereof configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure. In some examples, at least one processor and at least one memory coupled

with the at least one processor may be configured to perform one or more of the functions described herein (e.g., by one or more processors, individually or collectively, executing instructions stored in the at least one memory).

[0215] Additionally, or alternatively, the communications manager **1620**, the receiver **1610**, the transmitter **1615**, or various combinations or components thereof may be implemented in code (e.g., as communications management software or firmware) executed by at least one processor (e.g., referred to as a processor-executable code). If implemented in code executed by at least one processor, the functions of the communications manager **1620**, the receiver **1610**, the transmitter **1615**, or various combinations or components thereof may be performed by a general-purpose processor, a DSP, a CPU, an ASIC, an FPGA, a microcontroller, or any combination of these or other programmable logic devices (e.g., configured as or otherwise supporting, individually or collectively, a means for performing the functions described in the present disclosure).

[0216] In some examples, the communications manager **1620** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1610**, the transmitter **1615**, or both. For example, the communications manager **1620** may receive information from the receiver **1610**, send information to the transmitter **1615**, or be integrated in combination with the receiver **1610**, the transmitter **1615**, or both to obtain information, output information, or perform various other operations as described herein.

[0217] The communications manager **1620** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1620** is capable of, configured to, or operable to support a means for outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The communications manager **1620** is capable of, configured to, or operable to support a means for outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The communications manager **1620** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0218] By including or configuring the communications manager **1620** in accordance with examples as described herein, the device **1605** (e.g., at least one processor controlling or otherwise coupled with the receiver **1610**, the transmitter **1615**, the communications manager **1620**, or a combination thereof) may support techniques for reduced processing, reduced ambiguity for resource allocation for SBFD communications, and more efficient utilization of uplink and downlink communication resources based on per-BWP configuration of SBFD communications.

[0219] FIG. **17** shows a block diagram **1700** of a device **1705** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The device **1705** may be an example of aspects of a device **1605** or a network entity **105** as described herein. The device **1705** may include a receiver **1710**, a transmitter **1715**, and a communications manager **1720**. The device **1705**, or one or more components of the device **1705** (e.g., the receiver **1710**, the transmitter **1715**, the communications manager **1720**), may include at least one processor, which may be coupled with at least one memory, to support the described techniques. Each of these components may be in communication with one another (e.g., via one or more buses).

[0220] The receiver **1710** may provide a means for obtaining (e.g., receiving, determining, identifying) information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). Information may be passed on to other components of the device **1705**. In some

examples, the receiver **1710** may support obtaining information by receiving signals via one or more antennas. Additionally, or alternatively, the receiver **1710** may support obtaining information by receiving signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof.

[0221] The transmitter **1715** may provide a means for outputting (e.g., transmitting, providing, conveying, sending) information generated by other components of the device **1705**. For example, the transmitter **1715** may output information such as user data, control information, or any combination thereof (e.g., I/Q samples, symbols, packets, protocol data units, service data units) associated with various channels (e.g., control channels, data channels, information channels, channels associated with a protocol stack). In some examples, the transmitter **1715** may support outputting information by transmitting signals via one or more antennas. Additionally, or alternatively, the transmitter **1715** may support outputting information by transmitting signals via one or more wired (e.g., electrical, fiber optic) interfaces, wireless interfaces, or any combination thereof. In some examples, the transmitter **1715** and the receiver **1710** may be co-located in a transceiver, which may include or be coupled with a modem.

[0222] The device **1705**, or various components thereof, may be an example of means for performing various aspects of subband configuration techniques for SBFD as described herein. For example, the communications manager **1720** may include a BWP configuration component **1725**, an SBFD configuration component **1730**, an SBFD communication component **1735**, or any combination thereof. The communications manager **1720** may be an example of aspects of a communications manager **1620** as described herein. In some examples, the communications manager **1720**, or various components thereof, may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the receiver **1710**, the transmitter **1715**, or both. For example, the communications manager **1720** may receive information from the receiver **1710**, send information to the transmitter **1715**, or be integrated in combination with the receiver **1710**, the transmitter **1715**, or both to obtain information, output information, or perform various other operations as described herein.

[0223] The communications manager **1720** may support wireless communications in accordance with examples as disclosed herein. The BWP configuration component **1725** is capable of, configured to, or operable to support a means for outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The SBFD configuration component **1730** is capable of, configured to, or operable to support a means for outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The SBFD communication component **1735** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0224] FIG. **18** shows a block diagram **1800** of a communications manager **1820** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The communications manager **1820** may be an example of aspects of a communications manager **1620**, a communications manager **1720**, or both, as described herein. The communications manager **1820**, or various components thereof, may be an example of means for performing various aspects of subband configuration techniques for SBFD as described herein. For example, the communications manager **1820** may include a BWP configuration component **1825**, an SBFD configuration component **1830**, an SBFD communication component **1835**, a capability signaling management component **1840**, or any combination thereof. Each of these components, or components or subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses). The

communications may include communications within a protocol layer of a protocol stack, communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack, within a device, component, or virtualized component associated with a network entity **105**, between devices, components, or virtualized components associated with a network entity **105**), or any combination thereof.

[0225] The communications manager **1820** may support wireless communications in accordance with examples as disclosed herein. The BWP configuration component **1825** is capable of, configured to, or operable to support a means for outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The SBFD communication component **1835** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0226] In some examples, to support outputting the second information, the capability signaling management component **1840** is capable of, configured to, or operable to support a means for obtaining an indication of one or more capabilities of the UE. In some examples, to support outputting the second information, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting, responsive to the indication, the second information indicative of the SBFD configuration, where the SBFD configuration is based on the one or more capabilities of the UE.

[0227] In some examples, the SBFD configuration includes a resource indication value that indicates a resource allocation associated with the uplink subband, the downlink subband or both. In some examples, the resource indication value includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation. In some examples, the starting resource block is indicated relative to a first physical resource block of a carrier bandwidth. In some examples, the starting resource block is indicated relative to a first physical resource block of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE. In some examples, the uplink subband, the downlink subband, or both, are configured in an active BWP pair of the one or more BWP pairs. In some examples, the SBFD configuration is based on one or more capabilities of the UE, the one or more capabilities including a supported guard band size between the uplink subband and the downlink subband, a threshold size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.

[0228] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, where a resource block group size of the resource allocation is based on a size of the uplink subband, a size of the downlink subband, or both.

[0229] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, where the resource allocation of the scheduled physical channel includes a resource indication value that includes an indication of a starting resource block of the resource allocation that is with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.

[0230] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to

support a means for outputting an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, where the physical channel resource allocation indicates a starting resource block of each hop of the frequency hopping relative to a first resource block of the uplink subband, and where a resource block offset is based on a size of the uplink subband, the downlink subband, or both.

[0231] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting the second information indicative of the SBFD configuration in respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, where the SBFD configuration is indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration.

[0232] In some examples, the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more resource indication values, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof. In some examples, the SBFD configuration including the uplink subband, the downlink subband or both, is configured in accordance with a common resource block grid of a cell configured for SBFD communications. In some examples, the cell includes a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

[0233] In some examples, to support communicating the one or more messages, the SBFD communication component **1835** is capable of, configured to, or operable to support a means for communicating the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based on one or more physical resource blocks that at least partially overlap with the respective uplink BWP, the respective downlink BWP, or both, of the SBFD configuration.

[0234] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting, while the UE is in a connected state, the second information indicative of the SBFD configuration via a serving cell configuration message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0235] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting, while the UE is in a connected state, the second information indicative of the SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications.

[0236] In some examples, to support receiving the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting, while the UE is in a connected state, an idle state, or an inactive state, the second information indicative of the SBFD configuration via a cell common configuration of a broadcast message, where the SBFD configuration includes a time pattern or an indication of resources for SBFD communications. In some examples, outputting the SBFD configuration includes a SBFD pattern periodicity that is based on a TDD downlink and uplink periodicity.

[0237] In some examples, to support outputting the second information indicative of the SBFD configuration, the SBFD configuration component **1830** is capable of, configured to, or operable to support a means for outputting the second information indicative of the SBFD configuration via one or more broadcast system information messages, where the uplink subband, the downlink subband, or both, are configured relative to an initial BWP, are configured based on a serving cell bandwidth, or both.

[0238] FIG. 19 shows a diagram of a system 1900 including a device 1905 that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The device 1905 may be an example of or include components of a device 1605, a device 1705, or a network entity 105 as described herein. The device 1905 may communicate with other network devices or network equipment such as one or more of the network entities 105, UEs 115, or any combination thereof. The communications may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device 1905 may include components that support outputting and obtaining communications, such as a communications manager 1920, a transceiver 1910, one or more antennas 1915, at least one memory 1925, code 1930, and at least one processor 1935. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus 1940).

[0239] The transceiver 1910 may support bi-directional communications via wired links, wireless links, or both as described herein. In some examples, the transceiver 1910 may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some examples, the transceiver 1910 may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some examples, the device 1905 may include one or more antennas 1915, which may be capable of transmitting or receiving wireless transmissions (e.g., concurrently). The transceiver 1910 may also include a modem to modulate signals, to provide the modulated signals for transmission (e.g., by one or more antennas 1915, by a wired transmitter), to receive modulated signals (e.g., from one or more antennas 1915, from a wired receiver), and to demodulate signals. In some implementations, the transceiver 1910 may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas 1915 that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas 1915 that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver 1910 may include or be configured for coupling with one or more processors or one or more memory components that are operable to perform or support operations based on received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver 1910, or the transceiver 1910 and the one or more antennas 1915, or the transceiver 1910 and the one or more antennas 1915 and one or more processors or one or more memory components (e.g., the at least one processor 1935, the at least one memory 1925, or both), may be included in a chip or chip assembly that is installed in the device 1905. In some examples, the transceiver 1910 may be operable to support communications via one or more communications links (e.g., communication link(s) 125, backhaul communication link(s) 120, a midhaul communication link 162, a fronthaul communication link 168).

[0240] The at least one memory 1925 may include RAM, ROM, or any combination thereof. The at least one memory 1925 may store computer-readable, computer-executable, or processor-executable code, such as the code 1930. The code 1930 may include instructions that, when executed by one or more of the at least one processor 1935, cause the device 1905 to perform various functions described herein. The code 1930 may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some cases, the code 1930 may not be directly executable by a processor of the at least one processor 1935 but may cause a computer (e.g., when compiled and executed) to perform functions described herein. In some cases, the at least one memory 1925 may include, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some examples, the at least one processor 1935 may include multiple processors and the at least one memory 1925 may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories which may, individually or collectively, be

configured to perform various functions herein (for example, as part of a processing system).

[0241] The at least one processor **1935** may include one or more intelligent hardware devices (e.g., one or more general-purpose processors, one or more DSPs, one or more central processing units (CPUs), one or more graphics processing units (GPUs), one or more neural processing units (NPU) (also referred to as neural network processors or deep learning processors (DLPs)), one or more microcontrollers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some cases, the at least one processor **1935** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into one or more of the at least one processor **1935**. The at least one processor **1935** may be configured to execute computer-readable instructions stored in a memory (e.g., one or more of the at least one memory **1925**) to cause the device **1905** to perform various functions (e.g., functions or tasks supporting subband configuration techniques for SBFD). For example, the device **1905** or a component of the device **1905** may include at least one processor **1935** and at least one memory **1925** coupled with one or more of the at least one processor **1935**, the at least one processor **1935** and the at least one memory **1925** configured to perform various functions described herein. The at least one processor **1935** may be an example of a cloud-computing platform (e.g., one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (e.g., by executing code **1930**) to perform the functions of the device **1905**. The at least one processor **1935** may be any one or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the device **1905** (such as within one or more of the at least one memory **1925**). In some examples, the at least one processor **1935** may include multiple processors and the at least one memory **1925** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein. In some examples, the at least one processor **1935** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **1935**) and memory circuitry (which may include the at least one memory **1925**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **1935** or a processing system including the at least one processor **1935** may be configured to, configurable to, or operable to cause the device **1905** to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code stored in the at least one memory **1925** or otherwise, to perform one or more of the functions described herein.

[0242] In some examples, a bus **1940** may support communications of (e.g., within) a protocol layer of a protocol stack. In some examples, a bus **1940** may support communications associated with a logical channel of a protocol stack (e.g., between protocol layers of a protocol stack), which may include communications performed within a component of the device **1905**, or between different components of the device **1905** that may be co-located or located in different locations (e.g., where the device **1905** may refer to a system in which one or more of the communications manager **1920**, the transceiver **1910**, the at least one memory **1925**, the code **1930**, and the at least one processor **1935** may be located in one of the different components or divided between different components).

[0243] In some examples, the communications manager **1920** may manage aspects of communications with a core network **130** (e.g., via one or more wired or wireless backhaul links). For example, the communications manager **1920** may manage the transfer of data communications

for client devices, such as one or more UEs **115**. In some examples, the communications manager **1920** may manage communications with one or more other network entities **105**, and may include a controller or scheduler for controlling communications with UEs **115** (e.g., in cooperation with the one or more other network devices). In some examples, the communications manager **1920** may support an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities **105**.

[0244] The communications manager **1920** may support wireless communications in accordance with examples as disclosed herein. For example, the communications manager **1920** is capable of, configured to, or operable to support a means for outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs including a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The communications manager **1920** is capable of, configured to, or operable to support a means for outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, where the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The communications manager **1920** is capable of, configured to, or operable to support a means for communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0245] By including or configuring the communications manager **1920** in accordance with examples as described herein, the device **1905** may support techniques for improved communication reliability, reduced latency due to enhanced SBFD communications techniques and resource allocation, more efficient utilization of uplink and downlink communication resources in SBFD, improved coordination between devices, reduced ambiguity of resource allocation for SBFD uplink and downlink subbands, among other advantages.

[0246] In some examples, the communications manager **1920** may be configured to perform various operations (e.g., receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver **1910**, the one or more antennas **1915** (e.g., where applicable), or any combination thereof. Although the communications manager **1920** is illustrated as a separate component, in some examples, one or more functions described with reference to the communications manager **1920** may be supported by or performed by the transceiver **1910**, one or more of the at least one processor **1935**, one or more of the at least one memory **1925**, the code **1930**, or any combination thereof (for example, by a processing system including at least a portion of the at least one processor **1935**, the at least one memory **1925**, the code **1930**, or any combination thereof). For example, the code **1930** may include instructions executable by one or more of the at least one processor **1935** to cause the device **1905** to perform various aspects of subband configuration techniques for SBFD as described herein, or the at least one processor **1935** and the at least one memory **1925** may be otherwise configured to, individually or collectively, perform or support such operations.

[0247] FIG. **20** shows a flowchart illustrating a method **2000** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The operations of the method **2000** may be implemented by a UE or its components as described herein. For example, the operations of the method **2000** may be performed by a UE **115** as described with reference to FIGS. **1** through **15**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0248] At **2005**, the method may include receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs comprising a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The operations of **2005** may be performed in accordance with examples as disclosed

herein. In some examples, aspects of the operations of **2005** may be performed by a BWP configuration component **1425** as described with reference to FIG. **14**.

[0249] At **2010**, the method may include receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The operations of **2010** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2010** may be performed by an SBFD subband configuration component **1430** as described with reference to FIG. **14**.

[0250] At **2015**, the method may include communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration. The operations of **2015** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2015** may be performed by an SBFD communication component **1435** as described with reference to FIG. **14**.

[0251] FIG. **21** shows a flowchart illustrating a method **2100** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The operations of the method **2100** may be implemented by a UE or its components as described herein. For example, the operations of the method **2100** may be performed by a UE **115** as described with reference to FIGS. **1** through **15**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the described functions. Additionally, or alternatively, the UE may perform aspects of the described functions using special-purpose hardware.

[0252] At **2105**, the method may include receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs comprising a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The operations of **2105** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2105** may be performed by a BWP configuration component **1425** as described with reference to FIG. **14**.

[0253] At **2110**, the method may include transmitting an indication of one or more capabilities of the UE. The operations of **2110** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2110** may be performed by a UE capability signaling component **1440** as described with reference to FIG. **14**.

[0254] At **2115**, the method may include receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The operations of **2115** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2115** may be performed by an SBFD subband configuration component **1430** as described with reference to FIG. **14**.

[0255] At **2120**, the method may include receiving, responsive to the indication, the second information indicative of the SBFD configuration, where the SBFD configuration is based on the one or more capabilities of the UE. The operations of **2120** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2120** may be performed by an SBFD subband configuration component **1430** as described with reference to FIG. **14**.

[0256] At **2125**, the method may include communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration. The operations of **2125** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2125** may be performed by an SBFD communication component **1435** as described with reference to FIG. **14**.

[0257] FIG. **22** shows a flowchart illustrating a method **2200** that supports subband configuration techniques for SBFD in accordance with one or more aspects of the present disclosure. The

operations of the method **2200** may be implemented by a network entity or its components as described herein. For example, the operations of the method **2200** may be performed by a network entity as described with reference to FIGS. **1** through **11** and **16** through **19**. In some examples, a network entity may execute a set of instructions to control the functional elements of the network entity to perform the described functions. Additionally, or alternatively, the network entity may perform aspects of the described functions using special-purpose hardware.

[0258] At **2205**, the method may include outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs comprising a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE. The operations of **2205** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2205** may be performed by a BWP configuration component **1825** as described with reference to FIG. **18**.

[0259] At **2210**, the method may include outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs. The operations of **2210** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2210** may be performed by an SBFD configuration component **1830** as described with reference to FIG. **18**.

[0260] At **2215**, the method may include communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration. The operations of **2215** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **2215** may be performed by an SBFD communication component **1835** as described with reference to FIG. **18**.

[0261] The following provides an overview of aspects of the present disclosure:

[0262] Aspect 1: A method for wireless communications at a UE, comprising: receiving first information indicative of one or more BWP pairs configured for the UE, each BWP pair of the one or more BWP pairs comprising a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE; receiving second information indicative of a SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs; and communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0263] Aspect 2: The method of aspect 1, wherein receiving the second information comprises: transmitting an indication of one or more capabilities of the UE; and receiving, responsive to the indication, the second information indicative of the SBFD configuration, wherein the SBFD configuration is based at least in part on the one or more capabilities of the UE.

[0264] Aspect 3: The method of any of aspects 1 through 2, wherein the SBFD configuration includes a RIV that indicates a resource allocation associated with the uplink subband, the downlink subband or both, the RIV includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation.

[0265] Aspect 4: The method of aspect 3, wherein the starting resource block is indicated relative to a first PRB of a carrier bandwidth.

[0266] Aspect 5: The method of any of aspects 3 through 4, wherein the starting resource block is indicated relative to a first PRB of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE.

[0267] Aspect 6: The method of any of aspects 1 through 5, wherein the uplink subband, the downlink subband, or both, are configured in an active BWP pair of the one or more BWP pairs.

[0268] Aspect 7: The method of any of aspects 1 through 6, wherein the SBFD configuration is based at least in part on one or more capabilities of the UE, the one or more capabilities comprising a supported guard band size between the uplink subband and the downlink subband, a threshold

size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.

[0269] Aspect 8: The method of any of aspects 1 through 7, wherein the uplink subband corresponds to a respective virtual uplink BWP and the downlink subband corresponds to a respective virtual downlink BWP, and wherein receiving the second information indicative of the SBFD configuration comprises: receiving an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, wherein a resource block group size of the resource allocation is based at least in part on a size of the uplink subband, a size of the downlink subband, or both.

[0270] Aspect 9: The method of any of aspects 1 through 8, wherein the uplink subband corresponds to a respective virtual uplink BWP and the downlink subband corresponds to a respective virtual downlink BWP, and wherein receiving the second information indicative of the SBFD configuration comprises: receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, wherein the resource allocation of the scheduled physical channel comprises a RIV that includes an indication of a starting resource block of the resource allocation that is with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.

[0271] Aspect 10: The method of any of aspects 1 through 9, wherein the uplink subband corresponds to a respective virtual uplink BWP and the downlink subband corresponds to a respective virtual downlink BWP, and wherein receiving the second information indicative of the SBFD configuration comprises: receiving an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, wherein the physical channel resource allocation indicates a starting resource block of each hop of the frequency hopping relative to a first resource block of the uplink subband, and wherein a resource block offset is based at least in part on a size of the uplink subband, the downlink subband, or both.

[0272] Aspect 11: The method of any of aspects 1 through 10, wherein receiving the second information indicative of the SBFD configuration comprises: receiving the second information indicative of the SBFD configuration in respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, wherein the SBFD configuration is indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration.

[0273] Aspect 12: The method of aspect 11, wherein the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more RIVs, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

[0274] Aspect 13: The method of any of aspects 1 through 12, wherein the SBFD configuration including the uplink subband, the downlink subband or both, is configured in accordance with a common resource block grid of a cell configured for SBFD communications.

[0275] Aspect 14: The method of aspect 13, wherein the cell comprises a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

[0276] Aspect 15: The method of any of aspects 1 through 14, wherein communicating the one or more messages further comprises: communicating the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based at least in part on one or more PRBs that at least partially overlap with the respective uplink BWP, the respective downlink BWP, or both, of the SBFD configuration.

[0277] Aspect 16: The method of any of aspects 1 through 15, further comprising: dropping one or more PRBs of the respective uplink BWP, the respective downlink BWP, or both, based at least in

part on a subcarrier spacing of the respective uplink BWP, the respective downlink BWP, or both, being greater than a corresponding subcarrier spacing of the uplink subband, the downlink subband, or both.

[0278] Aspect 17: The method of any of aspects 1 through 16, wherein receiving the second information indicative of the SBFD configuration comprises: receiving, while the UE is in a connected state, the second information indicative of the SBFD configuration via a serving cell configuration message, wherein the SBFD configuration comprises a time pattern or an indication of resources for SBFD communications.

[0279] Aspect 18: The method of any of aspects 1 through 17, wherein receiving the second information indicative of the SBFD configuration comprises: receiving, while the UE is in a connected state, the second information indicative of the SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration comprises a time pattern or an indication of resources for SBFD communications.

[0280] Aspect 19: The method of any of aspects 1 through 18, wherein receiving the second information indicative of the SBFD configuration comprises: receiving, while the UE is in a connected state, an idle state, or an inactive state, the second information indicative of the SBFD configuration via a cell common configuration of a broadcast message, wherein the SBFD configuration comprises a time pattern or an indication of resources for SBFD communications.

[0281] Aspect 20: The method of any of aspects 1 through 19, wherein receiving the SBFD configuration comprises a SBFD pattern periodicity that is based at least in part on a TDD downlink and uplink periodicity.

[0282] Aspect 21: The method of any of aspects 1 through 20, wherein the UE is operating in an inactive state or an idle state, and receiving the second information indicative of the SBFD configuration comprises: receiving the second information indicative of the SBFD configuration via one or more broadcast system information messages, wherein the uplink subband, the downlink subband, or both, are configured relative to an initial BWP, are configured based at least in part on a serving cell bandwidth, or both.

[0283] Aspect 22: A method for wireless communications at a network entity, comprising: outputting first information indicative of one or more BWP pairs configured for a UE, each BWP pair of the one or more BWP pairs comprising a respective uplink BWP and a respective downlink BWP that at least partially occupy a channel bandwidth of the UE; outputting second information indicative of a SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more BWP pairs; and communicating one or more messages via the one or more BWP pairs in accordance with the SBFD configuration.

[0284] Aspect 23: The method of aspect 22, wherein outputting the second information comprises: obtaining an indication of one or more capabilities of the UE; and outputting, responsive to the indication, the second information indicative of the SBFD configuration, wherein the SBFD configuration is based at least in part on the one or more capabilities of the UE.

[0285] Aspect 24: The method of any of aspects 22 through 23, wherein the SBFD configuration includes a RIV that indicates a resource allocation associated with the uplink subband, the downlink subband or both, the RIV includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation.

[0286] Aspect 25: The method of aspect 24, wherein the starting resource block is indicated relative to a first PRB of a carrier bandwidth.

[0287] Aspect 26: The method of any of aspects 24 through 25, wherein the starting resource block is indicated relative to a first PRB of an active BWP of the one or more BWP pairs of the channel bandwidth of the UE.

[0288] Aspect 27: The method of any of aspects 22 through 26, wherein the uplink subband, the downlink subband, or both, are configured in an active BWP pair of the one or more BWP pairs.

[0289] Aspect 28: The method of any of aspects 22 through 27, wherein the SBFD configuration is based at least in part on one or more capabilities of the UE, the one or more capabilities comprising a supported guard band size between the uplink subband and the downlink subband, a threshold size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.

[0290] Aspect 29: The method of any of aspects 22 through 28, wherein the uplink subband corresponds to a respective virtual uplink BWP and the downlink subband corresponds to a respective virtual downlink BWP, and wherein outputting the second information indicative of the SBFD configuration comprises: outputting an indication of a resource allocation of a scheduled physical channel for SBFD communications via a bitmap, wherein a resource block group size of the resource allocation is based at least in part on a size of the uplink subband, a size of the downlink subband, or both.

[0291] Aspect 30: The method of any of aspects 22 through 29, wherein the uplink subband corresponds to a respective virtual uplink BWP and the downlink subband corresponds to a respective virtual downlink BWP, and wherein outputting the second information indicative of the SBFD configuration comprises: receiving an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, wherein the resource allocation of the scheduled physical channel comprises a RIV that includes an indication of a starting resource block of the resource allocation that is with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.

[0292] Aspect 31: The method of any of aspects 22 through 30, wherein the uplink subband corresponds to a respective virtual uplink BWP and the downlink subband corresponds to a respective virtual downlink BWP, and wherein outputting the second information indicative of the SBFD configuration comprises: outputting an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, wherein the physical channel resource allocation indicates a starting resource block of each hop of the frequency hopping relative to a first resource block of the uplink subband, and wherein a resource block offset is based at least in part on a size of the uplink subband, the downlink subband, or both.

[0293] Aspect 32: The method of any of aspects 22 through 31, wherein outputting the second information indicative of the SBFD configuration comprises: outputting the second information indicative of the SBFD configuration in respective uplink and downlink BWP configurations for the uplink subband, the downlink subband, or both, wherein the SBFD configuration is indicated jointly in each uplink and downlink BWP configuration, or separately for each uplink and downlink BWP configuration.

[0294] Aspect 33: The method of aspect 32, wherein the SBFD configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more RIVs, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

[0295] Aspect 34: The method of any of aspects 22 through 33, wherein the SBFD configuration including the uplink subband, the downlink subband or both, is configured in accordance with a common resource block grid of a cell configured for SBFD communications.

[0296] Aspect 35: The method of aspect 34, wherein the cell comprises a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

[0297] Aspect 36: The method of any of aspects 22 through 35, wherein communicating the one or more messages further comprises: communicating the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based at least in part on one or more PRBs that at least partially overlap with the respective uplink BWP, the respective downlink

BWP, or both, of the SBFD configuration.

[0298] Aspect 37: The method of any of aspects 22 through 36, wherein outputting the second information indicative of the SBFD configuration comprises: outputting, while the UE is in a connected state, the second information indicative of the SBFD configuration via a serving cell configuration message, wherein the SBFD configuration comprises a time pattern or an indication of resources for SBFD communications.

[0299] Aspect 38: The method of any of aspects 22 through 37, wherein outputting the second information indicative of the SBFD configuration comprises: outputting, while the UE is in a connected state, the second information indicative of the SBFD configuration for each of the one or more BWP pairs, wherein the SBFD configuration comprises a time pattern or an indication of resources for SBFD communications.

[0300] Aspect 39: The method of any of aspects 22 through 38, wherein receiving the second information indicative of the SBFD configuration comprises: outputting, while the UE is in a connected state, an idle state, or an inactive state, the second information indicative of the SBFD configuration via a cell common configuration of a broadcast message, wherein the SBFD configuration comprises a time pattern or an indication of resources for SBFD communications.

[0301] Aspect 40: The method of any of aspects 22 through 39, wherein outputting the SBFD configuration comprises a SBFD pattern periodicity that is based at least in part on a time domain duplexing downlink and uplink periodicity.

[0302] Aspect 41: The method of any of aspects 22 through 40, wherein outputting the second information indicative of the SBFD configuration comprises: outputting the second information indicative of the SBFD configuration via one or more broadcast system information messages, wherein the uplink subband, the downlink subband, or both, are configured relative to an initial BWP, are configured based at least in part on a serving cell bandwidth, or both.

[0303] Aspect 42: A UE for wireless communications, comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the UE to perform a method of any of aspects 1 through 21.

[0304] Aspect 43: A UE for wireless communications, comprising at least one means for performing a method of any of aspects 1 through 21.

[0305] Aspect 44: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by one or more processors to perform a method of any of aspects 1 through 21.

[0306] Aspect 45: A network entity for wireless communications, comprising one or more memories storing processor-executable code, and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the network entity to perform a method of any of aspects 22 through 41.

[0307] Aspect 46: A network entity for wireless communications, comprising at least one means for performing a method of any of aspects 22 through 41.

[0308] Aspect 47: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by one or more processors to perform a method of any of aspects 22 through 41.

[0309] It should be noted that the methods described herein describe possible implementations. The operations and the steps may be rearranged, or otherwise modified and other implementations are possible. Further, aspects from two or more of the methods may be combined.

[0310] Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and

Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

[0311] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0312] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed using a general-purpose processor, a DSP, an ASIC, a CPU, a graphics processing unit (GPU), a neural processing unit (NPU), an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor but, in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Any functions or operations described herein as being capable of being performed by a processor may be performed by multiple processors that, individually or collectively, are capable of performing the described functions or operations.

[0313] The functions described herein may be implemented using hardware, software executed by a processor, firmware, or any combination thereof. If implemented using software executed by a processor, the functions may be stored as or transmitted using one or more instructions or code of a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0314] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one location to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc. Disks may reproduce data magnetically, and discs may reproduce data optically using lasers. Combinations of the above are also included within the scope of computer-readable media. Any functions or operations described herein as being capable of being performed by a memory may be performed by multiple memories that, individually or collectively, are capable of performing the described functions or operations.

[0315] As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items

prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0316] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” and “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, the term “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” may refer to any or all of the one or more components. For example, a component introduced with the article “a” may be understood to mean “one or more components,” and referring to “the component” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.” Similarly, subsequent reference to a component introduced as “one or more components” using the terms “the” or “said” may refer to any or all of the one or more components. For example, referring to “the one or more components” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.”

[0317] The term “determine” or “determining” encompasses a variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database, or another data structure), ascertaining, and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data stored in memory), and the like. Also, “determining” can include resolving, obtaining, selecting, choosing, establishing, and other such similar actions.

[0318] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label or other subsequent reference label.

[0319] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some figures, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0320] The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A user equipment (UE), comprising: one or more memories storing processor-executable code; and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the UE to: receive first information indicative of one or more bandwidth part pairs configured for the UE, each bandwidth part pair of the one or more bandwidth part pairs comprising a respective uplink bandwidth part and a respective downlink bandwidth part that at least partially occupy a channel bandwidth of the UE; receive second information indicative of a subband full duplex configuration for each of the one or more bandwidth part pairs, wherein the subband full duplex configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more bandwidth part pairs; and communicate one or more messages via the one or more bandwidth part pairs in accordance with the subband full duplex configuration.
2. The UE of claim 1, wherein, to receive the second information, the one or more processors are individually or collectively operable to execute the code to cause the UE to: transmit an indication of one or more capabilities of the UE; and receive, responsive to the indication, the second information indicative of the subband full duplex configuration, wherein the subband full duplex configuration is based at least in part on the one or more capabilities of the UE.
3. The UE of claim 1, wherein the subband full duplex configuration includes a resource indication value that indicates a resource allocation associated with the uplink subband, the downlink subband or both, and the resource indication value includes an indication of a starting resource block of the resource allocation and a quantity of resource blocks included in the resource allocation, wherein the starting resource block is indicated relative to a first physical resource block of a carrier bandwidth, relative to a first physical resource block of an active bandwidth part of the one or more bandwidth part pairs of the channel bandwidth of the UE, or both.
4. The UE of claim 1, wherein the uplink subband, the downlink subband, or both, are configured in an active bandwidth part pair of the one or more bandwidth part pairs.
5. The UE of claim 1, wherein the subband full duplex configuration is based at least in part on one or more capabilities of the UE, the one or more capabilities comprising a supported guard band size between the uplink subband and the downlink subband, a threshold size of the uplink subband, a threshold size of the downlink subband, or any combination thereof.
6. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive an indication of a resource allocation of a scheduled physical channel for subband full duplex communications via a bitmap, wherein a resource block group size of the resource allocation is based at least in part on a size of the uplink subband, a size of the downlink subband, or both.
7. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive an indication of a resource allocation of a scheduled physical channel at the uplink subband, the downlink subband or both, wherein the resource allocation of the scheduled physical channel comprises a resource indication value that includes an indication of a starting resource block of the resource allocation that is with reference to a first resource block of the uplink subband or a first resource block of the downlink subband.
8. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive an indication of a physical channel resource allocation configured with frequency hopping within the downlink subband, the uplink subband, or both, wherein the physical channel resource allocation indicates a starting resource block of each hop of

the frequency hopping relative to a first resource block of the uplink subband, and wherein a resource block offset is based at least in part on a size of the uplink subband, the downlink subband, or both.

9. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive the second information indicative of the subband full duplex configuration in respective uplink and downlink bandwidth part configurations for the uplink subband, the downlink subband, or both, wherein the subband full duplex configuration is indicated jointly in each uplink and downlink bandwidth part configuration, or separately for each uplink and downlink bandwidth part configuration.

10. The UE of claim 9, wherein the subband full duplex configuration includes frequency location information for the uplink subband, the downlink subband, or both, one or more resource indication values, subcarrier spacing information, time domain information, semi-persistent scheduling information, one or more uplink channel configurations, one or more downlink channel configurations, one or more uplink reference signal configurations, one or more downlink reference signal configurations or any combination thereof.

11. The UE of claim 1, wherein the subband full duplex configuration including the uplink subband, the downlink subband or both, is configured in accordance with a common resource block grid of a cell configured for subband full duplex communications, wherein the cell comprises a network carrier and an associated network carrier bandwidth, a carrier of the UE and the channel bandwidth of the UE, or both.

12. The UE of claim 1, wherein, to communicate the one or more messages, the one or more processors are individually or collectively further operable to execute the code to cause the UE to: communicate the one or more messages via frequency resources of the uplink subband, the downlink subband, or both, based at least in part on one or more physical resource blocks that at least partially overlap with the respective uplink bandwidth part, the respective downlink bandwidth part, or both, of the subband full duplex configuration.

13. The UE of claim 1, wherein the one or more processors are individually or collectively further operable to execute the code to cause the UE to: drop one or more physical resource blocks of the respective uplink bandwidth part, the respective downlink bandwidth part, or both, based at least in part on a subcarrier spacing of the respective uplink bandwidth part, the respective downlink bandwidth part, or both, being greater than a corresponding subcarrier spacing of the uplink subband, the downlink subband, or both.

14. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive, while the UE is in a connected state, the second information indicative of the subband full duplex configuration via a serving cell configuration message, wherein the subband full duplex configuration comprises a time pattern or an indication of resources for subband full duplex communications.

15. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive, while the UE is in a connected state, the second information indicative of the subband full duplex configuration for each of the one or more bandwidth part pairs, wherein the subband full duplex configuration comprises a time pattern or an indication of resources for subband full duplex communications.

16. The UE of claim 1, wherein, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive, while the UE is in a connected state, an idle state, or an inactive state, the second information indicative of the subband full duplex configuration via a cell common configuration of a broadcast message, wherein the subband full duplex configuration

comprises a time pattern or an indication of resources for subband full duplex communications.

17. The UE of claim 1, wherein receiving the subband full duplex configuration comprises a subband full duplex pattern periodicity that is based at least in part on a time division duplexing downlink and uplink periodicity.

18. The UE of claim 1, wherein the UE is operating in an inactive state or an idle state, and, to receive the second information indicative of the subband full duplex configuration, the one or more processors are individually or collectively operable to execute the code to cause the UE to: receive the second information indicative of the subband full duplex configuration via one or more broadcast system information messages, wherein the uplink subband, the downlink subband, or both, are configured relative to an initial bandwidth part, are configured based at least in part on a serving cell bandwidth, or both.

19. A network entity, comprising: one or more memories storing processor-executable code; and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the network entity to: output first information indicative of one or more bandwidth part pairs configured for a user equipment (UE), each bandwidth part pair of the one or more bandwidth part pairs comprising a respective uplink bandwidth part and a respective downlink bandwidth part that at least partially occupy a channel bandwidth of the UE; output second information indicative of a subband full duplex configuration for each of the one or more bandwidth part pairs, wherein the subband full duplex configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more bandwidth part pairs; and communicate one or more messages via the one or more bandwidth part pairs in accordance with the subband full duplex configuration.

20. A method for wireless communications at a user equipment (UE), comprising: receiving first information indicative of one or more bandwidth part pairs configured for the UE, each bandwidth part pair of the one or more bandwidth part pairs comprising a respective uplink bandwidth part and a respective downlink bandwidth part that at least partially occupy a channel bandwidth of the UE; receiving second information indicative of a subband full duplex configuration for each of the one or more bandwidth part pairs, wherein the subband full duplex configuration indicates an uplink subband, a downlink subband or both, that corresponds to each of the one or more bandwidth part pairs; and communicating one or more messages via the one or more bandwidth part pairs in accordance with the subband full duplex configuration.
